Overlapping ownership in vertical relations: From fostering to tunneling

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The literature identifies several ways in which overlapping shareholders can reduce contractual and informational frictions between potential customers and suppliers. These effects, however, are compatible with overlapping owners also extracting private benefits from minority shareholders. I propose an instrumental variables approach to estimate the effect of more similar ownership structures separately from the effect of more concentrated ownership on either side of the relationship. My findings indicate that overlapping shareholders promote trading partnerships among companies within their portfolios, though the effect becomes four times larger when shareholder concentration differs substantially across companies. I argue that overlapping shareholders can create asymmetric incentives among managers, in which the diffuse-ownership company prioritizes the partner's gains while the concentrated company ignores potential losses for the other firm. Consistent with this mechanism, the same feature that makes a deal appealing to both parties can make it one-sided, fostering trading partnerships that enable tunneling along the supply chain.

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In the past, some have mistakenly assumed that our predominantly passive management style suggests a passive attitude with respect to corporate governance. Nothing could be further from the truth

William McNabb III
Vanguard's CEO, 2008-2018

The proliferation of large asset management institutions has accentuated the role of diversified minority shareholders in corporate strategy (Schmalz, 2018). In particular, with the rapid expansion of institutional financial intermediaries and a renewed preference for passive indexing strategies in the United States, a few institutional investors, such as Vanguard, BlackRock, and State Street, now rank among the top shareholders of most publicly listed companies in the country (Fichtner et al., 2017; Schmalz, 2018; Amel-Zadeh et al., 2022).[1]

The rise of large asset managers has led researchers to explore the implications of institutional investors holding stakes in two or more companies (henceforth, *overlapping shareholders*). As a result, a growing body of evidence now suggests that overlapping shareholders can soften competition (He and Huang, 2017; Azar et al., 2018), deter

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[1] Although the US stands out in the process, the trend is growing rapidly among large firms and certain geographic areas, such as Ireland, the Netherlands, and Sweden. Interestingly, recent evidence suggests that institutional ownership has been steadily high in regions like China, South America, and Mexico (Antón et al., 2025*b*).

entry (Xie and Gerakos, 2020; Newham et al., 2025), and produce supply-chain effects closely tied to the mechanisms of partial vertical integration, especially contractual and informational frictions (Chen, 2024; Deng et al., 2025; Freeman, 2025).

This paper builds on the finding that overlapping shareholders foster trading partnerships among companies in their portfolios, and asks whether they may use these partnerships to extract private benefits at the expense of non-overlapping shareholders. Contrary to previous studies, my analysis focuses on mechanisms of partial vertical integration associated with asymmetric managerial incentives. That is, when executives across companies align to favor a single firm, something that existing identification strategies cannot reliably capture.

My findings suggest that the likelihood of a trading partnership increases as ownership structures among potential customers and suppliers become more similar, but disproportionately when shareholder concentration differs—a pattern consistent with managers on the side with more diffuse ownership using these partnerships to extract private benefits.

Several theoretical and empirical studies analyze the connection between the ownership structure of strategically interacting firms and market outcomes.[2] However, much of the literature still assumes that managers maximize the value of the company that appointed them. While reasonable at first glance, this premise overlooks that shareholders—who appointed managers in the first place—prioritize the total value of their portfolios rather than the individual value of each firm they own. Therefore, whether through direct investor influence or because managers internalize other benefits from pleasing pivotal shareholders, a company's ownership structure can alter its goals, market strategies, and business relationships.

In light of this, researchers have looked into the impact of *cross-ownership* and *common ownership* in product markets. While the former describes settings in which two companies hold shares in each other, the latter denotes third-party institutional shareholders across close competitors. Evidence from several studies documents that common ownership can hinder competition in several industries (Azar et al., 2018; Torshizi and Clapp, 2021; Koch et al., 2021; Newham et al., 2025), to the point that the European Commission has begun invoking common ownership as an element of competitive context in merger decisions (Burnside and Kidane, 2021).

Meanwhile, the role of overlapping owners in vertical relations remains somewhat underexplored. Nonetheless, if overlapping shareholders drive executives to weigh externalities imposed on competitors, why would managers ignore effects across upstream and downstream markets?

The distinction between horizontal and vertically overlapping ownership also invites researchers to reinterpret canonical insights about the separation of ownership and control (Jensen and Meckling, 1976). The literature on overlapping ownership departs from the traditional corporate finance standpoint, in which managers may pursue private benefits rather than maximizing firm profits (Jensen, 1986). Instead, it focuses on the idea that executives maximize the joint returns of portfolios across the company's shareholders (Rotemberg, 1984).

Compared with horizontal settings, where companies and shareholders both benefit from softer competition, decisions along the supply chain can create winners and losers. Therefore, by tilting decisions toward the companies they hold most heavily, vertically overlapping shareholders would shift the axis of conflict—from managers versus owners

[2]For example, several theoretical contributions argue that partial stakes in rival companies soften competition by shifting firm objectives away from own profits (Reynolds and Snapp, 1986; Bresnahan and Salop, 1986; Reitman, 1994; Gilo et al., 2006). For empirical evidence on the mechanism, see, for example, Parker and Roller (1997); Alley (1997); Nain and Wang (2018); Heim et al. (2022).

to overlapping versus non-overlapping shareholders—so that the gains from a trading partnership need not accrue to the companies that decide on it.

The paper centers on publicly listed companies in the United States between 1999 and 2013, drawing on several open data sources. I rely on prior research papers that compiled public filings, which companies must periodically submit to the Securities and Exchange Commission (SEC). Ownership ties—covering large institutional investors and blockholders—come from 13D, 13F, and 13G filings (Backus et al., 2021; Schwartz-Ziv and Volkova, 2025), which I use to assemble diverse overlapping ownership metrics following the theoretical approach in Rotemberg (1984). Likewise, I identify potential suppliers from the vertical upstream relatedness metric in Frésard et al. (2020) and active trading partnerships from 10-K filings (Barrot and Sauvagnat, 2016).

The analysis uses two-way fixed-effects regressions to examine the relationship between ownership structures and trading partnerships, combining observational weights with an instrumental variable approach to address two empirical challenges. First, companies will never establish infeasible trading partnerships, regardless of the strength of the mechanisms. Second, portfolio decisions that create overlap and managerial decisions on trading partnerships occur simultaneously.

I account for potentially infeasible trading partnerships in my sample by weighting each customer-supplier dyad by its vertical upstream relatedness. That is, the extent to which a supplier's output serves as an input for the customer's business. This approach renders unnecessary any assumption about the supply chain structure across industries, while still down-weighting dyads that are unlikely to trade.

My analysis compares results across multiple fixed-effects specifications and uses instruments constructed from fifteen years of panel data on shareholdings to address omitted variable bias. These instruments mirror the overlapping ownership metrics used in the analysis but rely on predicted rather than observed shareholdings, so that the same procedure can be extended to any known formula of ownership structures. Contrary to prior identification strategies based on index changes, which rely on exclusion restrictions suited to unilateral decisions (e.g., Boller and Morton, 2020; Newham et al., 2025), my approach explicitly accounts for two-sided decisions, given that customers and suppliers jointly choose their trading partners.

Overall, increasing ownership-structure similarity by one standard deviation raises the likelihood of a trading partnership by roughly 10% to 19%—a magnitude comparable to Freeman (2025), who finds that overlapping owners reduce trading partnership terminations by roughly 26% to 32%. I examine the findings across a wide range of prediction models, sample definitions, and ownership metrics, and the conclusions are stable throughout.

Although prior studies contribute to the idea that overlapping shareholders help companies circumvent contractual and informational frictions (Riva, 2022; Chen, 2024; Freeman, 2025; Deng et al., 2025), the appendix includes empirical exercises using my identification strategy that suggest that holdup and double marginalization cannot fully explain the results. Furthermore, my work finds empirical patterns that hardly fit into that narrative: Overlapping owners make trading partnerships much more likely when a company has a more diffuse ownership structure than its counterpart.

My empirical findings indicate that ownership-structure similarity has disparate effects on the likelihood of a trading partnership, depending on the relative shareholder concentration across customers and suppliers. For instance, when customers and suppliers have close to identical shareholder concentration, increasing ownership-structure similarity by one standard deviation has no statistically significant effect on the likelihood of a trading partnership. By contrast, when shareholder concentrations differ substantially, the same effect ranges from 28% to 84%.

In my preferred specification, the effect roughly quadruples (from 19% to 84%) among customer-supplier dyads in which the relative shareholder concentration between the two companies deviates substantially from one. What drives such disproportionate effects when they stem from similar increases in ownership-structure similarity?

Throughout the paper, I argue that managers in firms with relatively diffuse ownership downplay outcomes within those firms, as gains and losses spread across many more minority shareholders. In other words, overlapping shareholders may bring about asymmetric managerial incentives, in which the diffusely owned company emphasizes its partner's gains. In contrast, the concentrated company ignores potential losses for the other firm. Ultimately, when decisions mostly benefit one side of the trading partnership, managers may pay closer attention to the company with more concentrated ownership, to the extent that the companies share investors.

My findings suggest that trading partnerships are more likely in situations where asymmetric managerial incentives are pervasive, creating opportunities for overlapping shareholders to extract private benefits along the supply chain, not necessarily by transferring equity value across companies but by cementing long-term advantages in upstream and downstream markets. In any case, managerial decisions may result in direct losses or foregone profits over time, to the detriment of non-overlapping shareholders at the company that has more diffuse ownership.

This paper contributes to the common ownership literature, whose theoretical foundations date back to Rotemberg (1984), in which firms tend to act collusively to favor shareholders with more diversified portfolios. Even so, the topic gained traction and drew widespread attention when numerous researchers started reporting that overlapping shareholders among close competitors can soften competition and lead to welfare losses. For example, previous studies find that common ownership drives higher market concentration (He and Huang, 2017) and prices (Azar et al., 2018; Park and Seo, 2019; Torshizi and Clapp, 2021), negative effects on employment and wages (Azar et al., 2025), and reduced market entry (Xie and Gerakos, 2020; Newham et al., 2025).

Still, several papers document positive aspects related to overlapping ownership, such as spillovers in research and development activities (López and Vives, 2019; Antón et al., 2025*a*), increased product variety (Aslan, 2024), or faster innovation diffusion (Kostovetsky and Manconi, 2020). In particular, the few studies on vertical relations stand out for outlining how overlapping shareholders alleviate contractual and informational frictions. For example, firms secure larger loans at lower interest rates or are more likely to obtain syndicated loans when shareholders overlap (Ojeda, 2018). Similarly, they increase the likelihood of companies joining strategic alliances (Lindsey, 2008), discourage suppliers from engaging in upward earnings management (Gao et al., 2022), boost upstream partner-specific investments (Chen, 2024; Deng et al., 2025), and extend the lifetime of existing trading partnerships (Freeman, 2025).

I add to this literature in two different ways. First, my empirical work offers suggestive evidence about a potential adverse mechanism: asymmetric managerial incentives resulting in executives favoring overlapping shareholders at the expense of non-overlapping ones. These findings add to the numerous studies warning of the risks of overlapping ownership. In fact, this body of evidence has led competition agencies to factor in common ownership in their decisions, though they have done so for horizontal rather than vertical mergers.[3]

[3]In the EU, the European Commission invoked common shareholdings as an element of competitive context. For instance, see the European Commission Decision of 27 March 2017 in Case COMP/M.7932—*Dow/DuPont*, particularly paragraphs 2338 to 2352, and the European Commission Decision of 21 March 2018 in Case COMP/M.8084—*Bayer/Monsanto*, particularly paragraphs 976 to 981 and 3303 to 3306. In the US, see the FTC–DOJ Statement of Interest of 22 May 2025 in *Texas et al. v. BlackRock, State Street, Vanguard*, where plaintiffs plead common holdings as a potential source of standalone harm.

Vertically overlapping ownership bears a close resemblance to partial vertical integration. Thus, prior studies have supported the view taken by regulatory bodies by placing greater emphasis on the mechanisms associated with efficiency gains. This paper, instead, brings up the drawbacks associated with partial vertical integration, arguing that overlapping ownership can lead to private benefits and tunneling opportunities worth considering.

My second contribution concerns the empirical strategy, which circumvents the limitations posed by other sources of exogenous variation. For example, Berger (2023) challenges fire sales from mutual fund scandals, while Lewellen and Lowry (2021) critique the use of financial institution mergers and index additions for identification. Although the literature has proposed other empirical strategies that work around earlier critiques, none address the inherent challenges of this setting.

To begin with, prior studies focus on market outcomes from unilateral company decisions, such as prices, entry, or investment, whereas trading partnerships involve two-sided decisions by customer and supplier managers. Implementing the same identification strategy in vertical settings would require shifts in the ownership structure not affecting other aspects of managerial decisions on either side of the supply chain relationship—a condition less likely to hold compared to settings with competing firms.

Likewise, prior identification strategies that exploit the addition of close competitors to indexes or ETFs (e.g., Antón et al., 2023; Newham et al., 2025; Azar et al., 2025) rely on systematic shifts in ownership-structure similarity across firms. While adequate for exploring symmetric effects on levels (such as market and firm outcomes), these empirical approaches fall short in capturing asymmetric managerial incentives. Identifying that variation requires an exogenous source of variation in shareholder concentration, which is what defines my design.

This paper also contributes to a broader literature about vertical relations. Studies typically look for answers in observed trading partnerships; to name a few examples, researchers have studied whether close relationships facilitate tax avoidance (Cen et al., 2017), how network centrality affects firm returns (Wu and Birge, 2014), whether vertical mergers can lead to foreclosure (Boehm and Sonntag, 2023), or how cross-ownership extends the lifespan of firms (Fee et al., 2006). By leveraging the vertical upstream relatedness between any two companies (Frésard et al., 2020), I avoid conditioning on the prior existence of the trading partnership. As a result, this paper incorporates the extensive margin into the analysis, extending the coverage to the set of feasible customer-supplier dyads. The applications of this methodology extend beyond my empirical analysis, providing a foundation for studying earlier questions on feasible rather than established trading partnerships.

The remainder of the paper proceeds as follows. Section I describes the data, sample, and metrics of overlapping ownership. Section II discusses the empirical approach, particularly the use of observational weights and the identification strategy. Section III shows that overlapping owners foster trading partnerships among companies within their portfolios, especially when asymmetric managerial incentives are pervasive. Finally, Section IV concludes.

I. Data: Ownership ties and the supply chain

This section describes the data sources, the sample, and the metrics I use throughout the main analysis. For alternative definitions used in the robustness empirical exercises, see Section C in the Appendix. I begin by introducing several open data sources on ownership and the supply chain and describing the sample construction. Next, I present two metrics that capture ownership patterns across companies: ownership-structure similarity (OSS) and relative shareholder concentration (RSC).

A. Data

The analysis centers on publicly listed companies in the United States from 1999 to 2013 excluding those in the financial sector (SIC codes 6000-6999) and the transportation, communication, and utility sectors (SIC codes 4000-4999).

Publicly listed companies. I use *Compustat North America – Fundamentals*, which provides quarterly firm-level data, including outstanding shares, market value, closing price, net sales, total assets, and research and development (R&D) investment. In addition, I compute the company age from the first report of book assets since 1976 and include the Herfindahl-Hirschman Index from the text-based network industry classification (TNIC-HHI) by Hoberg and Phillips (2016). Furthermore, I winsorize R&D investment and total assets at the 1st and 99th percentiles and adjust nominal values for inflation to 2012 US dollars using the implicit price deflator from the Bureau of Economic Analysis.

Ownership ties. The SEC requires asset managers holding over USD 100 million in North American securities to file Form 13F quarterly, regardless of the number of shares owned in a company. Blockholders, by contrast, must file Forms 13D and 13G once they acquire at least 5% of a company's outstanding shares, and annually thereafter.

For this paper, the US ownership network relies on two open data sources from prior research. I compile 13F filings submitted to the SEC between 1999 and 2017 from Backus et al. (2021) and combine them with 13D and 13G filings from 1998 to 2016 from Schwartz-Ziv and Volkova (2025).

Other SEC filings include Forms 3 to 5, which corporate insiders must file to inform ownership and transactions in company securities. They seldom coincide with institutional or blockholder positions, and are thus excluded from the analysis.

Trading partnerships. The supply chain network stems from the data published by Barrot and Sauvagnat (2016), covering trading partnerships among publicly listed companies in the United States and Canada from 1976 to 2013. Publicly traded companies file 10-K forms with the SEC every year and, under Statement of Financial Accounting Standards (SFAS) N°131, they must disclose the identity of any customer contributing at least 10% of total annual sales. This is a clear limitation in the data, as the wording of SFAS N°131 may lead to sample selection by making it less likely for small customers and large suppliers to exceed the mandatory 10% threshold.

In addition, *Compustat North America – Customer Segment* does not provide customer identifiers. Although prior studies have resorted to phonetic string-matching algorithms,[4] matching customers by disclosed names can introduce noise, as companies with similar names may refer to unrelated organizations despite shared historical roots.[5] Barrot and Sauvagnat (2016) extend the procedure by combining a phonetic string-matching algorithm with manual identification, which allows them to recover a more accurate structure of the supply chain network.

Vertical upstream relatedness. I use the *vertical upstream relatedness* across company dyads to identify potential trading partners in the data. This directed measure appears in Frésard et al. (2020) and captures the degree to which a company produces outputs that another company uses as inputs.

The procedure to obtain this metric involves mapping product descriptions from 10-K filings to a broad range of US commodities. The authors combine this information

[4]See, for example, Fee et al. (2006), Wu and Birge (2014), or Cheung et al. (2020).

[5]For example, A.T. Massey Coal Company spun off Massey Energy Co. in 2000 and changed its name to Fluor Corp. Therefore, the string "Massey Coal Company" should match "Fluor Corp." rather than "Massey Energy Co.," which most phonetic string-matching algorithms would miss.

with input-output intensities across all commodities, so they obtain a set of candidate companies that produce the required inputs for every listed product in 10-K filings. Depending on the number of products and the input-output intensities across them, the procedure yields directed measures of upstream relatedness for every pair of publicly listed US companies.[6] Along with their paper, the authors have published annual updates from 1989 to 2023, which I employ to identify the subset of feasible customer-supplier dyads among all possible combinations of two companies.

In practice, values close to one are extremely rare, as they would imply that every output of the supplier serves as an input to every good the customer produces. Thus, most company dyads have values close to zero, meaning that a trading partnership among them is practically unfeasible.

B. Sample

The sample draws on the data sources described above, which overlap from 1999 to 2013. Panel A in Table 1 shows that roughly 7,500 companies are in business every year during this period, along with other relevant cross-year average characteristics.

However, the analysis covers a shorter window, spanning from 2000 to 2013. This restriction arises from my identification strategy, which requires a one-year lag in each company's ownership structure. For further details on the empirical approach, see Section II.

Moving forward, the paper focuses on the relationship between customers and suppliers, so that Panels B to E in Table 1 distinguish between companies that appear as customers or suppliers in the data. Nonetheless, many publicly listed companies buy upstream and sell downstream, and therefore appear in both roles.

Feasible trading partners. Feasibility in trading partnerships hinges on what companies produce, not on whether they have an established relationship. For instance, Nokia announced in 2011 that it would ditch its flagship operating system, Symbian, and license Windows 7 and Bing from Microsoft. However, the agreement suggests that Microsoft could have been a potential supplier to Nokia even before 2011.

The challenge, though, is to capture all feasible trading partnerships, regardless of whether I observe them at any point in time. Otherwise, the analysis would assume technological constraints that make trading partnerships infeasible, when in reality it could reflect the endogenous choices of customers and suppliers.

In contrast, consider Activision Blizzard Inc., a video game holding company, and Alexion Pharmaceuticals Inc., a pharmaceutical and biotechnology company. Although both were publicly traded between 2005 and 2013, it is no surprise that the two companies did not establish a trading partnership, given the industries they operate in and the commodities they produce.

Infeasible customer-supplier dyads can bias my results because, regardless of their ownership structures, their products make them incompatible trading partners.[7] As a first step, I keep dyads with positive vertical upstream relatedness or an observed trading partnership.[8] Subsection II.B describes how using vertical upstream relatedness as observational weights helps address the bias, and Subsection C.1 in the Appendix shows that the conclusions remain unchanged when excluding trading partnerships that are supposedly infeasible but observed in the data.

[6] Notice that customer and supplier identities matter in this setting, so each company pair spans two different customer-supplier dyads.

[7] The direction of this bias is ambiguous: it depends on whether ownership-structure similarity is higher among feasible or infeasible dyads, which in turn determines whether OLS estimates are biased upward or downward.

[8] The distinction stems from the fact that Frésard et al. (2020) lose data when mapping CUSIP (American Bankers Association) to GVKEY (Compustat) identifiers. Thus, supposedly infeasible customer-supplier dyads still appear in Compustat's Customer Segment.

Table 1 – *Cross-year summary statistics for companies in Compustat*

	N	3-digit SIC industries	Age	Market value	Annual sales	Total assets	R&D intensity	TNIC-HHI	MB ratio
Panel A — Compustat									
All companies	7,374	222	11.44	2,854.45	2,329.67	1,860.12	26.00%	3,221	-0.41
Panel B — Feasible trading partners									
Customers	3,440	212	15.57	4,161.67	3,589.36	2,599.80	12.23%	3,224	2.65
Suppliers	3,472	213	15.40	3,499.50	2,770.85	2,213.27	12.45%	3,223	2.62
Panel C — Trading partnerships									
Customers	408	104	20.48	24,931.11	21,921.18	13,159.84	6.22%	2,563	3.25
Suppliers	889	132	15.26	3,618.89	2,332.86	2,137.56	13.24%	3,078	2.52
Panel D — Disclosing suppliers									
Customers	355	92	20.74	27,662.17	24,381.00	14,308.44	6.36%	2,520	3.29
Suppliers	664	112	15.92	3,649.98	2,377.55	2,140.80	13.21%	2,975	2.35
Panel E — Non-disclosing suppliers									
Customers	177	66	21.95	40,559.70	36,015.65	18,721.08	4.75%	2,588	3.65
Suppliers	225	77	13.49	3,491.98	2,221.38	2,107.96	13.61%	3,390	2.79

Notes: The table reports cross-year summary statistics for publicly listed US companies from 2000 to 2013. Panel A describes companies in industries other than the financial, transportation, communication, and utility sectors. Panels B and C focus on feasible customer-supplier dyads and trading partnerships observed in the data. Panel D restricts to suppliers that have disclosed at least one customer below the 10% threshold since 1976, along with their respective customers. Panel E, instead, centers on non-disclosing suppliers and their respective customers. The first and second columns indicate the average number of companies each year and the average number of 3-digit SIC industries they span, respectively. Market value, annual sales, and total assets are in millions of 2012 US dollars.

The final sample contains 164,853,088 annual observations from 2000 to 2013, comprising 28,551,658 unique feasible trading partnerships between 6,018 customers and 6,113 suppliers. As mentioned above, the feasible set is drawn from 10-K product descriptions mapped to commodities, so companies with broader product descriptions are more likely to match with a counterparty. Consistent with this, customers and suppliers in Panel B in Table 1 have longer lifespans and lower R&D intensity, as well as higher sales, total assets, market values, and market-to-book ratios than the average firm in the Compustat dataset.

Trading partnerships. Following Barrot and Sauvagnat (2016), I define customer-supplier dyads as trading partnerships throughout all years between the first and last time the supplier reports the other firm as a significant customer in a 10-K filing. This procedure yields 23,723 trading partnerships between 2000 and 2013, comprising 5,823 unique dyads between 1,023 customers and 2,226 suppliers. Notice that trading partnerships are rare within the set of feasible customer-supplier dyads—roughly one in every seven thousand—and reported results must account for the small magnitudes.

Despite the wording of SFAS N°131, large suppliers do not appear underrepresented in the Customer Segment, given that cross-year average supplier characteristics do not differ significantly between Panels B and C. The problem arises among small customers because they make relatively smaller purchases and are less likely to surpass the 10% threshold for mandatory disclosure. Comparing cross-year average customer characteristics across Panels B and C shows that customers in the Customer Segment have a lifespan nearly 5 years longer, lower engagement in R&D activities, and significantly higher sales, market value, and total assets.[9]

[9]For further review, see Barrot and Sauvagnat (2016) and Wu and Birge (2014). Both papers compare the Compustat–Customer Segment with alternative data sources, such as Capital IQ and Bloomberg SPLC.

Differences are more striking among customers purchasing from suppliers that appear stricter with the Statement. Although there is no obligation, many companies voluntarily disclose customers below the mandatory threshold. I classify suppliers by whether they have disclosed a customer below the 10% threshold at least once since 1976 (Panel D) or not (Panel E), with both panels displaying cross-year average characteristics for suppliers and their respective customers.

Panels B to E describe a problematic selection in the trading partnerships I observe in the Customer Segment. Section II details my empirical strategy and how I address these concerns in its design.

C. *Ownership-structure similarity and relative shareholder concentration*

Overlapping ownership enters the analysis through two features associated with the interaction between ownership and corporate decisions: ownership-structure similarity and relative shareholder concentration across companies. In Rotemberg (1984), managers maximize a weighted average of the portfolios held by all the shareholders in the firm. The model highlights that investors do not need to actively engage with managerial decisions, but can still influence executives through perceived control and stock ownership. For a more detailed description of the framework and its derivation, see Section A in the Appendix.

The model's derivations show that maximizing the average value of portfolios held by the firm's shareholders is equivalent to maximizing the weighted sum of profits across all firms in the economy, where the non-negative *profit weight* κ_{jkt} denotes how the manager in company j weighs the effects of their decisions on the profits of company k in year t. Namely,

$$\kappa_{jkt} = \overbrace{\cos(\beta_{jt}, \beta_{kt})}^{\substack{\text{Ownership-structure} \\ \text{similarity (OSS)}}} \cdot \underbrace{\frac{\|\beta_{kt}\|}{\|\beta_{jt}\|}}_{\substack{\text{Relative shareholder} \\ \text{concentration (RSC)}}} \tag{1}$$

where β_{jt} depicts the ownership structure of company j in year t.[10] In addition, Equation (1) shows that κ_{jkt} can be written as the product of two interpretable terms that constitute the core of my analysis.

The first term in Equation (1) represents *ownership-structure similarity* (OSS), defined as the cosine similarity between the ownership structures of companies j and k, which ranges from 0 to 1.[11] A higher value in OSS indicates greater overlap among shareholders and more balanced ownership. For instance, consider two investors who initially hold shares in different companies. If they both sold their shares to a third investor with no prior holdings in either of them, the increase would reflect greater ownership overlap. By contrast, consider two overlapping shareholders, each with a stronger position in one company. If they exchanged shares proportionally to their portfolios, OSS would rise as shareholdings become more balanced.

The second term in Equation (1) captures *relative shareholder concentration* (RSC) across companies, defined as the ratio of the Euclidean norms of the corresponding ownership structures. The term gains its name because both the numerator and denominator match the square root of the Herfindahl-Hirschman Index of investor concentration for companies k and j, respectively. Thus, a higher shareholder concentration

[10] Note that β_{jt} is an $|\mathcal{I}|$-dimensional vector with elements in $[0, 1]$, representing the fraction of total shares held by investor $i \in \mathcal{I}$ in firm $j \in \mathcal{J}$. Since minority and insider shareholdings are unobserved, the elements of β_{jt} do not necessarily add up to 1.

[11] Cosine similarity is the complement of the angular distance between two vectors in an inner product space, and it characterizes whether they point in roughly the same direction.

in the company that appoints the manager reduces RSC. Whereas a higher shareholder concentration in the firms whose profits are weighted increases RSC.

OSS is a shape-only (scale-free) statistic that determines whether managers care about the other company's profits at all. Without shared investors, profit weights become zero, and managers turn indifferent to the externalities they may impose on potential trading partners.

RSC, by contrast, is a scale-only (shape-free) statistic that determines whether managerial attention is symmetrical. Managers assign identical profit weights to each other only when shareholder concentration is the same across the two firms, regardless of OSS. However, a company whose owners are more concentrated can capture a disproportionate share of attention from its partner's manager, which is the exact mechanism through which companies with diffuse ownership may be taken advantage of.

This reasoning finds a correlate in previous work. For instance, Backus et al. (2021) uses the framework in Rotemberg (1984) and concludes that profit weights above one imply tunneling incentives, as defined by Johnson et al. (2000), among executives. Along the same lines, Gilje et al. (2020) develop a metric for overlapping ownership that incorporates asymmetric managerial incentives through the internalization of voting rather than cash flow rights.

I combine all ownership ties from 13D, 13F, and 13G filings—excluding duplicated stakes where an investor files both 13F and 13D/G—and reconstruct the ownership network for US publicly listed companies between 1999 and 2013. Each entry indicates the fraction of outstanding shares owned by investor $i \in \mathcal{I}$ in company $j \in \mathcal{J}$. Following Equation (1), I employ these data to construct annual metrics of OSS and RSC, which I use throughout the main analysis. For a discussion about alternative metrics found in the literature and the corresponding robustness analysis, see Subsection C.2.

II. Empirical strategy

In this section, I define the empirical approach used throughout the paper. I begin by describing the empirical exercises that constitute the core of my analysis and the three fixed-effects specifications that help me account for the selection bias discussed in Subsection I.B. The following subsections detail the empirical challenges that arise from unobserved infeasibility and endogeneity, as well as my strategies to address them. I describe how observational weights, based on vertical upstream relatedness among dyads, can account for infeasible trading partnerships that remain in the sample. Second, I discuss my instrumental variables approach, which uses analogous versions of OSS and RSC based on predicted rather than observed shareholdings.

A. Estimating fostering and tunneling effects

Two empirical exercises sit at the core of the paper: overlapping shareholders foster trading partnerships, and the effect strengthens when shareholder concentration differs significantly between customers and suppliers. I estimate both effects using panel data regressions and a linear probability model, comparing estimates across different specifications to assess potential confounders when drawing conclusions.

Fostering. The first empirical exercise studies whether overlapping shareholders foster trading partnerships among companies in their portfolios. For that reason, I estimate the following empirical model

$$\text{LTP}_{sct} \;=\; \delta_{FE} \;+\; \tau \cdot \text{OSS}_{sct} \;+\; \varepsilon_{sct} \tag{2}$$

where LTP_{sct} denotes the *likelihood of a trading partnership*, defined by whether supplier s and customer c had a trading partnership during year t, and δ_{FE} represents

one of the three fixed-effects specifications described below, each one controlling for different sources of potential biases.

Given that trading partnerships are rare within the set of feasible customer-supplier dyads, I normalize LTP and OSS to facilitate interpretation. Therefore, τ reports the percentage change in the average probability of a trading partnership in the sample (roughly 0.02%) associated with a one-standard-deviation increase in OSS. I compute both averages using observational weights, as discussed in the next subsection.

Furthermore, I follow Angrist and Pischke (2009) and cluster standard errors at the dyad-level, where both LTP and OSS are measured. Importantly, the sample contains tens of millions of dyads, so the cluster-robust estimator does not suffer from small-cluster problems.

Tunneling. To explore the role of asymmetric managerial incentives, I extend the empirical model described in Equation (2) to account for any differences in shareholder concentration across customers and suppliers. Namely,

$$\text{LTP}_{sct} \;=\; \delta_{FE} \;+\; \tau_1 \cdot \text{OSS}_{sct} \;+\; \tau_2 \cdot \ln(\text{RSC}_{max}) \;+\; \tau_3 \cdot \text{OSS}_{sct} \times \ln(\text{RSC}_{max}) \;+\; \varepsilon_{sct} \quad (3)$$

where RSC_{max} quantifies the distance in shareholder concentrations across companies, defined as the larger of the two ratios, $\max(\text{RSC}_{sct}, \text{RSC}_{cst})$.

A benefit of this specification is that τ_1 captures the effect of ownership overlap separately from differences in shareholder concentration, which is captured by τ_2. Furthermore, the framework predicts that RSC should affect managerial incentives only to the extent that companies share investors. In the absence of overlapping shareholders, managers should not internalize differences in shareholder concentration. This specification provides a direct test for the prediction: τ_2 should be statistically insignificant. By contrast, asymmetric managerial incentives should operate exclusively through the interaction effect, which τ_3 captures separately from τ_2.

Backus et al. (2021) find that RSC (*relative investor concentration* in their paper) explains between one-third and one-half of the total variation in profit weights. Arguably, the apparent rationale is that the variable represents an unbounded ratio with a close-to-Pareto distribution. This feature justifies using the log-ratio of RSC in the estimation, as well as dropping dyads with companies with extremely diffuse ownership, for which RSC_{max} tends to infinity. Thus, I keep observations with RSC within one standard deviation of the mean, which covers roughly 71% of the original sample and 90% of all dyads in which both companies have at least one institutional shareholder or blockholder.

As in the fostering empirical exercise, I normalize LTP, OSS, and RSC to simplify coefficient interpretation, and cluster standard errors at the dyad level.

The literature recognizes multiple reasons why certain firms are more likely to form trading partnerships (Bernard et al., 2019; Carvalho and Tahbaz-Salehi, 2019), and Table 1 shows that customers in observed trading partnerships tend to be larger companies. Combined with the censoring introduced by the wording of SFAS N°131, OSS and RSC estimates could pick up variation unrelated to the mechanisms I propose. To account for this, I develop three fixed-effects specifications, each addressing a different source of bias.

Time-varying company-FE ($\delta_{st} + \delta_{ct}$)**.** A natural response to potential confounders is to add time-varying covariates to the empirical model, such as firm age, total assets, annual sales, and industry concentration. This path, however, leaves the problem unresolved and introduces another potential identification threat. To begin with, it does nothing about unobserved firm characteristics associated with the likelihood of

trading partnerships. In addition, many covariates face the same simultaneity problems as ownership structure, and would act as *bad controls* when added directly to the empirical model (Angrist and Pischke, 2009).

The first specification includes time-varying company fixed effects, controlling for idiosyncratic trends across companies. It resembles two-way fixed effects (TWFE) estimation, while taking advantage of the fact that I can split the unit of analysis—the dyad—into customers and suppliers. Thus, the estimation still captures variation in the outcome and ownership across potential trading partners, as every company in the sample has multiple feasible customers and suppliers each year.

Dyad-FE and year-FE ($\delta_{sc} + \delta_t$)**.** This combination of fixed effects follows a conventional TWFE specification, with one flexibly controlling for aggregate time trends and the other accounting for time-invariant idiosyncratic characteristics at the unit-of-analysis level. The upside is that dyad-FE holds constant several unobserved predictors of trading partnerships in the data, such as geographic proximity, relative company size, and other determinants associated with SFAS N°131 censoring. The downside is that the estimation captures only within-dyad variation in the outcome and ownership. Since the mechanisms I discuss insinuate that most variation lies between dyads, the coefficients from this specification may underestimate the effects.

Dyad-FE and time-varying company-FE ($\delta_{sc} + \delta_{st} + \delta_{ct}$)**.** The third specification combines the previous two, capturing within-dyad variation across customers and suppliers of every company. Thus, it is the most stringent of the three, accounting for most potential confounders but leaving the least variation for estimation.

Altogether, the three fixed-effects specifications complement each other. The first specification—my preferred set of coefficients throughout the analysis—produces estimates that better capture variation across dyads, though it cannot account for potential confounders in the association between customers and suppliers. The other two specifications move in the opposite direction, controlling more flexibly for potential confounders but capturing only within-dyad variation. As such, these specifications likely understate the magnitude of the coefficients, while still supporting the general direction of the effects and interpretations about the role of overlapping shareholders along the supply chain.[12]

B. Using weights to account for infeasible trading partnerships

Trading partnerships represent only a tiny fraction of feasible customer-supplier dyads, suggesting that the sample may still contain many unlikely trading partners. In this subsection, I discuss why infeasible dyads can bias my estimates and how observational weights—proportional to vertical upstream relatedness—can alleviate these concerns.

Unlike previous studies, I avoid conditioning the sample on the prior existence of a trading partnership, allowing me to incorporate the extensive margin into the analysis.[13] That is, studying not only whether the presence of overlapping shareholders affects trading partnerships already in place but also whether they promote new links along the supply chain.

[12] The first and third specifications absorb firm-year error components, so the remaining dependence, if any, lies within the dyad over time, which dyad-level clusters capture appropriately. Conversely, the second specification overlaps with the clustering strategy, meaning that confidence intervals do not account for error dependencies across customers and suppliers and can undercover the true sampling variation.

[13] For example, Fee et al. (2006) and Freeman (2025) focus on supply chain relationships already in place and, as such, can only explore intensive margins, such as sales, or the decision to extend the partnership for an additional year.

However, populating the sample with every two-firm combination introduces omitted-variable bias, given that whether the two companies can engage in a trading partnership at all remains unobserved. Moreover, restricting the sample to customer-supplier dyads with strictly positive vertical upstream relatedness may not be enough, as the issue still arises, though more subtly.[14]

Let me illustrate the problem and a potential solution with an example. For that, consider a simpler version of Equation (2)

$$\text{LTP}_{sct} = \alpha \ + \ \tau \cdot \text{OSS}_{sct} \ + \ X_{sct} \cdot \beta \ + \ \varepsilon_{sct}$$

in which I replace the fixed effects with some arbitrary set of controls X_{sct}. Suppose the empirical model is correctly specified for feasible trading partnerships, so the WLS estimates for (α, τ, β) that originate from minimizing the following expression

$$\mathbb{E}\Big[\big(\text{LTP}_{sct} = \alpha \ + \ \tau \cdot \text{OSS}_{sct} \ + \ X_{sct} \cdot \beta\big)^2 \omega_{sct}\Big]$$

are unbiased and consistent, where ω_{sct} represent arbitrary observational weights that enter the expression multiplicatively.

In addition, let $\pi\left(F_{sct} = 1\right) \in [0, 1]$ denote the inherent probability that a given trading partnership is feasible, which the econometrician cannot directly observe. When $F_{sct} = 0$, companies cannot engage in trading partnerships that are ultimately infeasible (like the one between Activision Blizzard and Alexion Pharmaceuticals from the example in Subsection I.A), in which case $\text{LTP}_{sct} = 0$. Therefore, I can express the expected error in the estimation as follows

$$\begin{aligned}\mathbb{E}\Bigg[\mathbb{E}\left[\left(m_{sct} \ - \ f_{sct}\right)^2 \omega_{sct} \middle| F_{sct} = 1\right] \pi\left(F_{sct} = 1\right) \\ + \mathbb{E}\left[\left(\alpha \ + \ \tau \cdot \text{OSS}_{sct} \ + \ \beta \cdot X_{sct}\right)^2 \omega_{sct} \middle| F_{sct} = 0\right] \Big(1 - \pi\left(F_{sct} = 1\right)\Big)\Bigg]\end{aligned}$$

where m_{sct} and f_{sct} represent the conditional mean and the best predicting function, respectively, when $F_{sct} = 1$ and the customer-supplier dyad is feasible.

Because the estimation minimizes only the first term, failing to account for the feasibility of the customer-supplier dyad can lead to sizable, systematic bias from the second term. Moreover, as $\pi\left(F_{sct} = 1\right)$ approaches zero, the estimator's target becomes irrelevant for predicting the likelihood of a trading partnership.

A first approach would be to model the determinants that make the customer-supplier dyad infeasible explicitly. For instance, by incorporating adequate controls into X_{sct}, it would be possible to offset the bias in the second term without affecting the predictive power from the first term. Unfortunately, such controls are often unavailable, and I must resort to other means.

A second approach is to drop dyads more aggressively when they are suspected to be infeasible, as they would systematically contribute to the bias in the second term. Setting a higher vertical upstream relatedness threshold for including customer-supplier dyads can help alleviate the problem, given that the highest probabilities of that occurring ensue when $\pi\left(F_{sct} = 1\right)$ is close to zero. Nonetheless, while incrementally

[14] One way to think about this limitation is that no company in the sample manufactures a single good or offers a single service, as publicly listed companies in the US tend to be multiproduct firms. Therefore, a supplier needs only to produce at least one commodity the customer uses to produce at least one of its outputs for vertical upstream relatedness to yield a positive but near-zero value. Indeed, the problem spreads as inputs or goods signify a tiny contribution to the whole production process.

raising the threshold further reduces concerns, it can eventually lead me to discard unlikely—though still feasible—dyads from the sample.

A third approach, and the one I follow, is to set weights ω_{sct} that tend to zero when the likelihood of a feasible customer-supplier dyad is low. So far, I have assumed that a null vertical upstream relatedness is informative enough about infeasibility. Therefore, under the same assumption, I can use the same variable as observational weights, downweighting the second term toward zero and significantly reducing bias.

C. Identification strategy: Predicted shareholdings

Investors and managers make portfolio and trading-partnership decisions around the same time, so my empirical strategy requires an exogenous source of variation that also works in vertical settings, where customers and suppliers jointly decide on trading partnerships. This subsection details the steps to construct instruments for OSS and RSC, using predicted rather than observed shareholdings to construct analogous versions of the same metrics.

The literature on common ownership has paid special attention to endogeneity problems associated with using ownership-related measures to explain outcomes at the firm- or market-level (Schmalz, 2018, 2021). Nevertheless, a few examples should suffice to illustrate the same concerns in vertical settings.

Active investors pay close attention to the supply chain before committing to investment decisions, casting doubt on the direction of the mechanisms due to reverse causality. Similarly, imagine a company makes a revolutionary technological innovation. The company may enter new trading partnerships to source the inputs required to produce the new technology. At the same time, a newly granted patent could attract the attention of several informed investors, increasing their holdings not only in the innovator but also in companies they believe might benefit from the novelty upstream and downstream. This example, along with other unobservable factors, can simultaneously affect ownership and trading partnerships, leading to omitted-variable bias.

Despite the initial back-and-forth over valid strategies for addressing endogeneity in ownership structures, a few papers document exogenous sources of variation that effectively work around the problem (e.g., Boller and Morton, 2020; Antón et al., 2023; Newham et al., 2025; Azar et al., 2025).[15] However, these studies use strategies based on index changes whose exclusion restriction relies on the unilateral nature of the decisions they study—such as prices or entry—and do not extend to vertical settings, where customers and suppliers jointly establish trading partnerships. The problem is that index changes shift the ownership structure of at least one company in the dyad and, at the same time, cannot be tied to managerial decisions in other unrelated companies.[16]

The alternative I propose builds on the idea that sufficiently lagged values of a covariate can be valid instruments, provided the outcome's persistence decays faster than the covariate's (Anderson and Hsiao, 1981). Yet my identification strategy takes a turn. Instead of using lagged values of OSS and RSC, I construct analogous versions of these metrics using the predictions from a first-order Markovian process estimated on fifteen years of institutional shareholdings.

[15] For the initial debate and exchange of opinions, see Kennedy et al. (2017), Azar et al. (2018), Dennis et al. (2022), and Azar et al. (2022). Other examples include critiques in Berger (2023), who describes problems associated with fire sales resulting from mutual fund scandals, and Lewellen and Lowry (2021), who discusses the limitations of financial institution mergers and additions to indexes.

[16] Several papers use index changes as exogenous sources of variation. Boone and White (2015), Kennedy et al. (2017), Brooks et al. (2018), and Kostovetsky and Manconi (2020) use Russell Index reconstitutions, while Aghion et al. (2013) use S&P 500 additions. The novelty in Boller and Morton (2020) lies in not considering the outcomes of the added companies themselves, but those of competitors, for whom the changes in index constituents are completely unrelated.

This methodology allows me to build instruments for any known function of shareholdings, which is the novel contribution for metrics like RSC. So far, identification strategies relied on shifts in the average OSS across firms and markets, produced by the addition of close competitors into indices and ETFs.

Predicted shareholdings. For the main analysis, I use the following first-order Markovian process for predicting shareholdings

$$\beta_{ijt} \;=\; \alpha \;+\; \rho \cdot \beta_{ijt-1} \;+\; \nu_{ijt} \tag{4}$$

where β_{ijt} denotes the fraction of outstanding shares owned by investor $i \in \mathcal{I}$ on company $j \in \mathcal{J}$ in year t.

I argue that the residual variation, ν_{ijt}, captures most of the endogeneity that biases OLS estimates, assuming the chosen number of lags (one for this specification) satisfies the exclusion restriction. In other words, as long as the supply chain persistence decays faster than institutional shareholdings persistence. Subsection C.3 in the Appendix reports a robustness analysis and shows that alternative specifications of the prediction model and using longer lags do not change the paper's conclusions. If anything, using fewer lags stands as the conservative choice.

Given that the model uses one-year lagged shareholdings, the estimation automatically excludes institutional owners in years when they first purchased shares in a company. This becomes a byproduct of the instrumental variables approach, but contributes to the interpretation that compliers belong to the portfolio of shareholders with fewer perturbations in their holdings—as these companies will tend to have a stronger first stage.[17] Subsection C.4 in the Appendix extrapolates the intuition to companies with more stable ownership structures. After excluding companies in years when they participated in, or were related to, S&P 500 index changes, I find no statistically significant differences in estimates.

I use OLS to estimate Equation (4) on the same ownership network described in Section I. The ownership network comprises panel data on annual ownership ties between US companies and institutional shareholders or blockholders, spanning 1999 to 2013. The estimation yields the following coefficients

$$\begin{array}{cccccc} \alpha & 0.0005 & \rho & 0.9075 & R^2 & 0.814 \\ & (0.000010) & & (0.002219) & & \end{array}$$

all of which are statistically significant at the 99% confidence level when using standard errors robust to heteroskedasticity. More importantly, the R^2 is 0.814, suggesting that past shareholdings explain a sizable portion of the variation in portfolios.

Predicted and observed ownership structures differ, as estimates from Equation (4) do not fully capture sudden portfolio shifts. Thus, my approach treats these holdings as if they were in the hands of retail shareholders—individual investors who buy and sell company shares in small amounts—when computing ownership structures.

I feed the predicted ownership structures back into the formulas described in Subsection I.C. This procedure yields analogous versions of OSS and RSC that I use as instruments, muting the endogenous response of investment decisions without sacrificing the non-linearities embedded in the overlapping ownership metrics. Furthermore, I build similar instruments for a few alternative overlapping ownership metrics in the literature, which I cover in detail in Subsection C.2 in the Appendix.

[17] Because both covariates and their analogous instruments are continuous, the 2SLS estimand (LATE) corresponds to the average effects across compliers, weighted proportionally by its first-stage marginal effect (Angrist et al., 2000).

As discussed above, the exclusion restriction entails a crucial identification assumption. As such, fixed effects must effectively absorb supply chain persistence in both steps of the 2SLS estimation to keep estimates unbiased. The first and third specifications—which include time-varying customer and supplier fixed effects—are more likely to capture company-specific trends and control for persistence driven by specific customers and suppliers. Likewise, the second and third specifications—which include dyad fixed effects—control for supply chain persistence driven by specific trading partners. Together, these considerations reinforce the idea that all three specifications complement each other in interpreting the results.

A potential argument against my instrument is that supply chain persistence depends on idiosyncratic, time-varying characteristics of the trading partnership itself. Unfortunately, such dependence coincides with the identifying variation and is therefore beyond the scope of my empirical strategy.

III. Overlapping shareholders: Turning vertical relations into private benefits?

Whereas overlapping shareholders may foster trading partnerships among companies in their portfolios, the gains from these vertical relations need not be shared equally, turning many trading partnerships into one-sided vertical relations over time. This section presents empirical evidence that overlapping shareholders promote trading partnerships between companies in their portfolios, and that effects strengthen when shareholder concentration differs substantially across companies.

My results align with previous research showing that ownership overlap increases the likelihood of active trading partnerships. While these studies argue that overlapping shareholders foster trading partnerships by reducing contractual and informational frictions, this paper focuses on the role of asymmetric managerial incentives. Nevertheless, Section B in the Appendix presents additional empirical exercises that speak directly to other results in the literature, such as holdup (Deng et al., 2025; Freeman, 2025) and double marginalization (Riva, 2022). Overall, the results are consistent with these mechanisms but suggest that they cannot fully account for the empirical patterns.

A. Overlapping shareholders foster trading partnerships

This subsection presents empirical evidence that the likelihood of an active trading partnership increases with OSS between customers and suppliers, suggesting that overlapping shareholders foster trading partnerships among companies in their portfolios.

For this empirical exercise, I estimate Equation (2) using OLS and 2SLS, as described in Section II. Two distinct sources of bias can distort OLS estimates for OSS. First, measurement error in ownership structures—likely due to unobserved insider and retail holdings—creates attenuation bias that pushes coefficients toward zero. Second, omitted-variable bias can shift estimates in either direction, depending on whether unaccounted determinants of trading partnerships positively or negatively correlate with ownership.

Figure 1 reports the coefficients across specifications, which indicate that a one-standard-deviation increase in OSS raises the average probability of a trading partnership by 9.92% to 19.78%, depending on the fixed-effects specification. For details, see Table SA.1 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages.

The three specifications convey a single message: overlapping shareholders in feasible customer-supplier dyads foster trading partnerships, despite differences in magnitude. For instance, my preferred specification (orange) points to around a 20% increase in the average probability, while the ones that include dyad fixed effects (blue and green) settle

for roughly a 10% increase. Considering that the later specifications exploit only within-dyad variation, the combined results strengthen the case that overlapping ownership fosters trading partnerships rather than merely reflecting between-dyad selection.

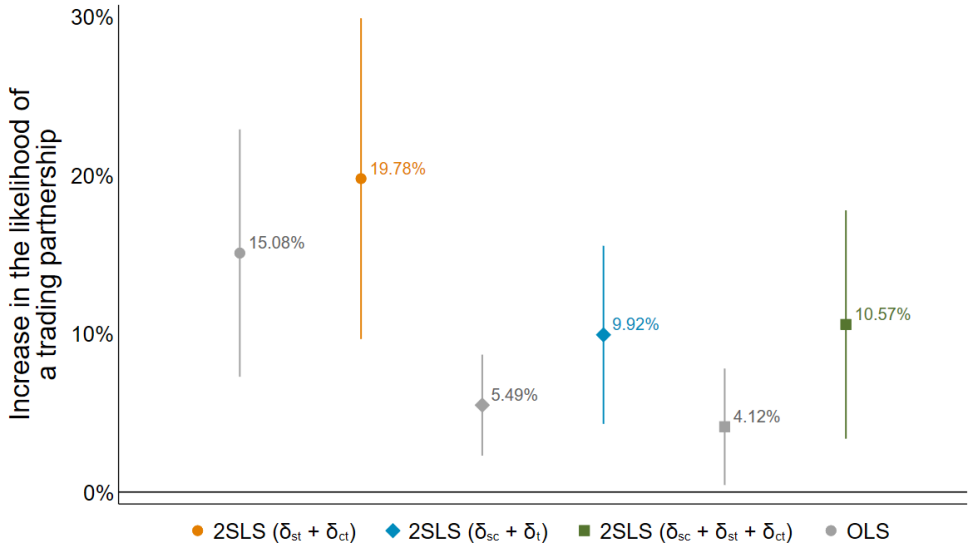
Notably, 2SLS coefficients are slightly higher, despite being statistically indistinguishable from their OLS counterparts. Freeman (2025) reaches the same conclusion using financial institutional mergers as their main identification strategy. So, 2SLS either alleviates a seemingly modest attenuation bias or handles biases that offset each other, in which case a positive bias of comparable magnitude exists, e.g., reverse causality.

I repeat the empirical exercise using a second-order Markovian process for predicting ownership structures. The comparison across the two prediction models implies a negative correlation between ownership and the unaccounted determinants of trading partnerships, meaning the instrumental variables approach accounts for combined, though modest, attenuation and omitted-variable bias. For a comparison of the results across prediction models, see Subsection C.3 in the Appendix.

Building on this interpretation, 2SLS estimates report local average treatment effects (LATE), and complier dyads in my setting involve companies with more stable ownership structures over time—such as passive index investors and institutional shareholders with long-term stakes. Thus, the similarity between OLS and 2SLS estimates may also shed light on the mechanism, emphasizing which shareholders are most likely to influence managerial decisions.

Arguably, long-standing investors have the most influence over managers: not necessarily because they directly persuade executives, but because managers tend to cater

Figure 1 – *Ownership-structure similarity (OSS) increases the likelihood of trading partnerships*



Notes: The figure depicts the effect of a one-standard-deviation increase in OSS on the average likelihood of a trading partnership. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I define OSS in Equation (1). I estimate Equation (2) using OLS (left, gray) and 2SLS (right, color) across three FE specifications, as described in Subsection II.A: (customer×year)-FE and (supplier×year)-FE (orange circle), dyad-FE and year-FE (blue diamond), and both combined (green square). The 2SLS instrument is an analogous version of OSS, based on predicted ownership structures, as described in Subsection II.C. Confidence intervals report dyad-cluster robust standard errors at the 95% confidence level. All regressions drop singleton observations under the most stringent FE specification (green square). For further reference, see Subsection III.A for a description of the analysis and Table SA.1 in the Appendix for details.

to pivotal shareholders with longer investment horizons. Thus, managers weigh more heavily how their choices affect other companies in those investors' portfolios.

So far, the findings establish that overlapping shareholders encourage trading partnerships among companies in their portfolios; whether those business relationships yield symmetric gains for customers and suppliers is a separate question. The next subsection presents suggestive evidence that the same ownership overlap can create asymmetric managerial incentives, leading to tunneling opportunities along the supply chain.

B. Tunneling: Privately benefiting from trading partnerships

This subsection shows that the baseline effect of increasing OSS strengthens when a company has significantly more diffuse ownership than its trading partner. Inspired by certain features of partial vertical integration, where shareholders may use the supply chain for their private benefit, I argue that my empirical findings are suggestive of companies preferring trading partnerships with asymmetric managerial incentives.

For this empirical exercise, I estimate Equation (3) using OLS and 2SLS, as described in Section II. For details, see Table SA.2 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages.

The framework I use to derive OSS and RSC in Subsection I.C establishes that RSC is a scale-only statistic, meaning that OSS is the only component that contains information about the shape of profit weights. In other words, RSC should not directly affect the outcome, only through OSS.

Therefore, when reporting the results, I find it more transparent to convey their interaction using the following expression for the heterogeneous marginal effects

$$\tau(\text{RSC}_{max}) \;=\; \tau_1 \;+\; \tau_3 \cdot \ln(\text{RSC}_{max})$$

rather than the interaction coefficient alone. Thus, $\tau(\text{RSC}_{max})$ reports the marginal effect of a one-standard-deviation increase in OSS, holding the distance in shareholder concentration across companies, $\ln(\text{RSC}_{max})$, constant.

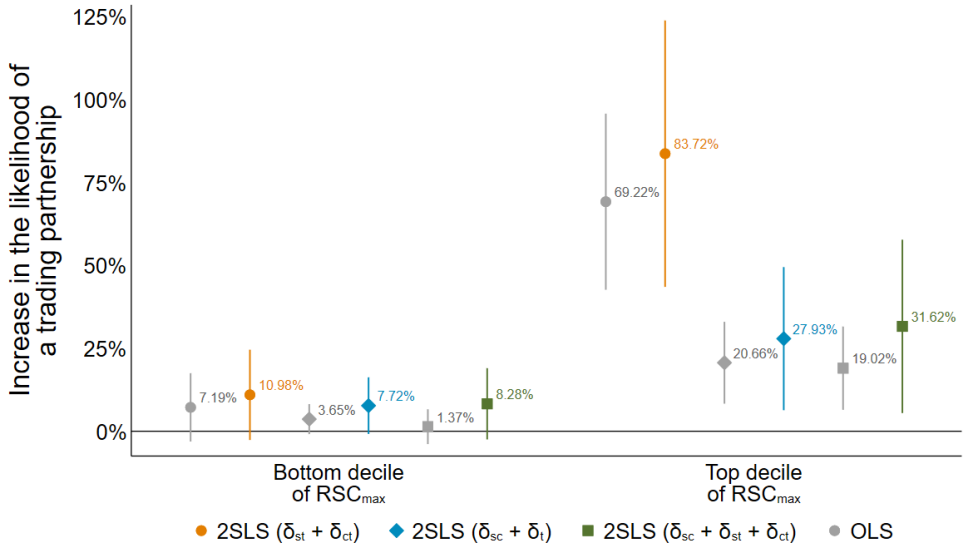
Figure 2 displays the marginal effects across estimations. Coefficients on the left report marginal effects at the bottom decile of the RSC_{max} distribution (1.0889), while coefficients on the right do so at the top decile (5.2089). In this way, the figure separates the margin on which managers decide to attend to a trading partner's profits at all (bottom decile, nearly identical concentration) from the margin on which that attention becomes asymmetric (top decile, strong asymmetric managerial incentives).

Coefficients on the left indicate that an increase in OSS has no statistically significant effect on the likelihood of a trading partnership when RSCs are close to one. That is, when managers in the customer-supplier dyad assign almost identical profit weights to each other's companies. By contrast, the coefficients on the right indicate that a one-standard-deviation increase in OSS raises the average probability of a trading partnership by 27.93% to 83.72%, depending on the fixed-effects specification, roughly quadrupling fostering estimates.

The combined results offer an alternative reading of results in other empirical papers, as ownership overlap does not appear to drive the results in Subsection III.A on its own. Studies that use single composite metrics, such as *overlapping institutional ownership* (Gao et al., 2022; Chen, 2024; Deng et al., 2025; Freeman, 2025), conflate the shape and scale margins of overlapping ownership. Therefore, the mechanisms in those studies may respond to the asymmetric-concentration margin rather than overlapping shareholders alone.

The comparison, however, indicates that managers are more inclined to establish and extend trading partnerships in which a party has a relatively diffuse ownership.

Figure 2 – *Relative shareholder concentration (RSC) amplifies the effect of ownership-structure similarity (OSS)*



Notes: The figure depicts the marginal effect of a one-standard-deviation increase in OSS on the average likelihood of a trading partnership for dyads in the bottom (1.0889) and top (5.2089) deciles of RSC. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I keep dyads with RSC within one standard deviation of the mean, as described in Subsection II.A. I define OSS and RSC in Equation (1). I estimate Equation (3) using OLS (left, gray) and 2SLS (right, color) across three FE specifications, as described in Subsection II.A: (customer×year)-FE and (supplier×year)-FE (orange circle), dyad-FE and year-FE (blue diamond), and both combined (green square). The 2SLS instruments are analogous versions of OSS and RSC, based on predicted ownership structures, as described in Subsection II.C. Confidence intervals report dyad-cluster robust standard errors at the 95% confidence level. All regressions drop singleton observations under the most stringent FE specification (green square). For further reference, see Subsection III.B for a description of the analysis and Table SA.2 in the Appendix for details.

Although the empirical exercise remains silent on the exact channel, the framework underlying OSS and RSC metrics offers an interpretable mechanism.

Intuitively, the model conveys that trading partnerships become more likely when managers weigh the externalities imposed on the potential trading partner more heavily. In particular, when the trading partner's manager does not reciprocate. Next, I argue that the empirical patterns align with situations in which managerial decisions enable tunneling opportunities from companies with diffuse ownership to those with concentrated ownership. Overlapping shareholders can then privately benefit from these opportunities and, in the process, harm non-overlapping shareholders at the diffuse-ownership firm that appointed the conflicted manager.

Tunneling. Figure 2 presents suggestive evidence that companies prefer trading partnerships with asymmetric managerial incentives. The baseline effect of increasing OSS roughly quadruples in customer-supplier dyads where a company has significantly more diffuse ownership than its trading partner.

Notably, trading partnerships exhibit a strikingly higher likelihood when the framework anticipates that managers weigh gains with trading partners more heavily than losses at the companies that appointed them. In other words, for situations in which overlapping shareholders have the opportunity to influence managerial decisions and use the supply chain for their private benefit, negatively affecting the diffuse-owned company and, in particular, its non-overlapping shareholders. Meanwhile, other efficiency-

gain mechanisms, including holdup and double marginalization, fall short in justifying these empirical patterns.

The first mechanism that comes to mind when overlapping shareholders privately benefit from managerial decisions at the expense of other investors is *tunneling* (Johnson et al., 2000; Atanasov et al., 2007). That is, when a company transfers assets or cash flows to another.

For example, consider a group of overlapping shareholders who own 60% of the equity in customer c and 20% of the equity in supplier s, and suppose these shareholders can prompt managers to set a lower input price than otherwise would be the case. Then, for every USD 100 million the customer saves on input costs, overlapping shareholders would pocket a net difference of USD 40 million. Furthermore, non-overlapping shareholders in the downstream company would benefit as well, obtaining another USD 40 million, while non-overlapping shareholders in the upstream company would lose the entire USD 80 million.

However, as Ehrhardt and Nowak (2015) explains, tunneling takes many forms and does not necessarily imply monetary transfers. Overlapping shareholders may pursue other goals, such as extending their control over firms, the supply chain, or the degree of competition across markets.

As such, tunneling is analogous to many of the mechanisms associated with partial vertical integration, which the literature has documented and discussed as a separate issue. For example, Boehm and Sonntag (2023) finds evidence that vertical mergers increase the likelihood of foreclosure, and Newham et al. (2025) show that pharmaceuticals can deter the entry of generics. Similarly, Levy et al. (2018) develop a theoretical framework in which companies can acquire partial stakes in vertically related companies to foreclose rivals, which can be profitable, depending on the ownership and control structures of the parties involved.

The framework in Rotemberg (1984) offers a broad interpretation that connects the implications of tunneling, partial vertical integration, and overlapping ownership. The model posits that managers assign a positive profit weight to a company with whom ownership overlaps, with Backus et al. (2021) going so far as to label profit weights above one as tunneling, given that managers would strictly prioritize the weighted firm over the one that appointed them. Nevertheless, smaller profit weights affect managerial decisions, especially in vertical settings, where asymmetric incentives can shift how companies are willing to split the gains from doing business together.

All in all, overlapping ownership has implications similar to those of partially integrated companies, which can, in turn, create tunneling opportunities—in a broad sense. That is, not only through transfers of equity value and profits across companies, but also through decisions that leave overlapping shareholders better off than they otherwise would be.

SFAS N°131—Censoring as a lower bound. A natural concern is that marginal effects may reflect the censoring introduced by SFAS N°131 rather than a genuine economic mechanism. As discussed in Subsection I.A, the Statement requires disclosure only when a customer accounts for at least 10% of a supplier's annual sales. Therefore, small customers and large suppliers are mechanically less likely to appear in the Customer Segment.

To test this further, I split the sample by the side of the trading partnership with the more concentrated ownership structure. If censoring were driving the results, one would expect significant differences across subsamples. First, trading partnerships with diffuse-ownership suppliers would exhibit smaller sales ratios to meet the SFAS N°131 threshold. Second, disclosure rates would differ across subsamples. Third, customer and supplier shareholder concentrations would drastically differ across subsamples.

The comparison shows none of these patterns. First, sales ratios on trading partnerships are roughly 100 times larger than on other customer-supplier dyads, but are around the same order of magnitude across subsamples. Second, disclosure rates are nearly identical across subsamples, with around 77% of suppliers disclosing at least one customer below the 10% threshold since 1976. Third, customer and supplier shareholder concentrations are symmetric across subsamples.

The descriptives confirm censoring, but they do not support it as the source of asymmetric managerial incentives. Moreover, the two specifications that include dyad fixed effects should absorb most of the remaining association between the threshold and RSC.

Nevertheless, unobserved trading partnerships between large suppliers and small customers can still bias my estimates toward zero. Because the sales ratio in customer-supplier dyads likely correlates with RSC, censoring disproportionately zeros out trading partnerships where the supplier has diffuse ownership, and the customer is concentrated—the very dyads where the empirical model predicts tunneling effects to be strongest.

Therefore, I estimate Equation (3) separately on each subsample. For details, see Table SA.3 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages. Notice that by splitting the sample, time-varying company fixed effects fully absorb RSC, so it is dropped from the estimation.

Figure 3 supports the intuition but also suggests caution when speaking to the magnitudes of these effects. For instance, Panel (a) reports coefficients solely for dyads where customers have more diffuse ownership. While the conclusions align with those of Figure 2, the magnitudes under asymmetric managerial incentives are systematically larger, suggesting that earlier results may constitute a lower bound on the true effects. The estimates report that a one-standard-deviation increase in OSS raises the average probability of a trading partnership by 89.32% to 164.10%, depending on the fixed-effects specification.

In contrast, Panel (b) reports coefficients only for dyads where suppliers have more diffuse ownership, which are harder to interpret individually because of censoring. Taken at face value, the point estimates suggest that RSC reduces the effect of increasing OSS when customers are more concentrated. However, as the outcome in this subsample contains much more noise, confidence intervals are larger, and coefficients are not statistically distinguishable from zero.

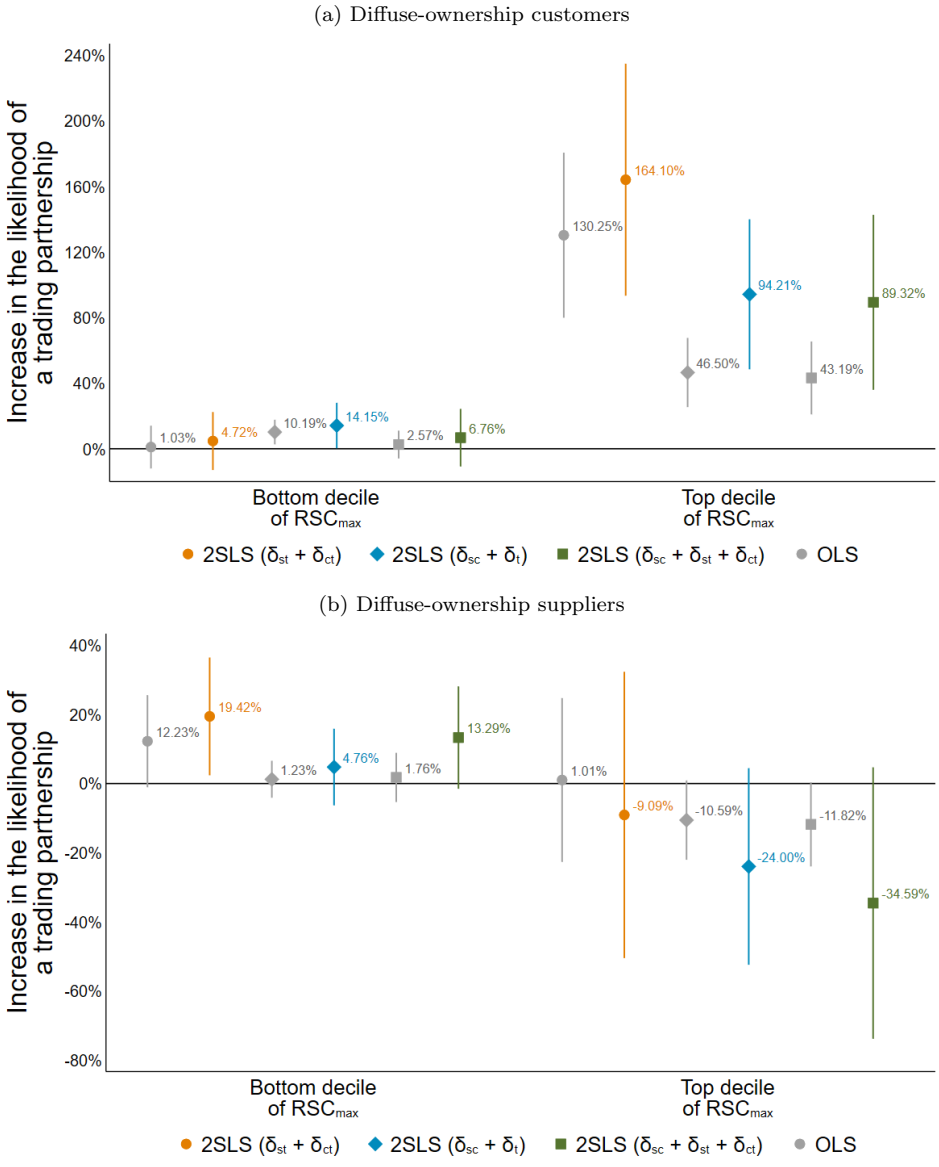
In conclusion, the asymmetric-concentration margin appears larger than the previous empirical exercise suggested. However, the unobserved trading partnerships in the data coincide with the very dyads where expected tunneling effects are strongest. Therefore, at this point, it is impossible to anticipate where the diffusion mechanism is truly symmetric or whether it responds to economic mechanisms solely on the customer side of the relationship.

IV. Conclusion

This paper contributes to the growing literature on common ownership. While early work has shown that overlapping shareholders can influence managerial decisions to soften competition among rivals (e.g., Azar et al., 2018; Torshizi and Clapp, 2021; Newham et al., 2025), I ask whether the same mechanism operates when companies lie on opposite sides of a potential trading partnership.

Recent studies on vertically overlapping ownership emphasize the role of institutional shareholders in mitigating contractual and informational frictions (Chen, 2024; Freeman, 2025; Deng et al., 2025), which support the idea that overlapping ownership and partial vertical integration share several features and mechanisms.

Figure 3 – *Tunneling effects are stronger when customers have more diffuse ownership*



Notes: The figure depicts the marginal effect of a one-standard-deviation increase in *ownership-structure similarity* (OSS) on the average likelihood of a trading partnership for dyads in the bottom and top deciles of *relative shareholder concentration* (RSC), depending on whether customers or suppliers have a relatively diffuse ownership structure. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I split the analysis by the side of the trading partnership with the more diffuse ownership structure. Panels (a) and (b) report coefficients solely for dyads where customers or suppliers have more diffuse ownership, respectively. I keep dyads with RSC within one standard deviation of the mean, as described in Subsection II.A. I define OSS and RSC in Equation (1). I estimate Equation (3) separately on each subsample, using OLS (left, gray) and 2SLS (right, color) across three FE specifications, as described in Subsection II.A: (customer×year)-FE and (supplier×year)-FE (orange circle), dyad-FE and year-FE (blue diamond), and both combined (green square). The 2SLS instruments are analogous versions of OSS and RSC, based on predicted ownership structures, as described in Subsection II.C. Confidence intervals report dyad-cluster robust standard errors at the 95% confidence level. All regressions drop singleton observations in each subsample under the most stringent FE specification (green square). For further reference, see Subsection III.B for a description of the analysis and Table SA.3 in the Appendix for details.

My work takes a different route and focuses on mechanisms analogous to tunneling (Johnson et al., 2000). By combining several open data sources from 1999 to 2013 on US ownership and supply chain networks, I find that overlapping shareholders in customer-supplier dyads increase the likelihood of trading partnerships. Interestingly, this effect strengthens when a trading partner has a relatively more diffuse ownership structure, amplifying the baseline effect by a factor of four. In other words, the effect strengthens when the underlying framework anticipates that managers appointed by companies with diffuse ownership become more inclined to transfer resources out of the company toward other potential trading partners with more concentrated shareholdings. These results also hold across alternative sample definitions, ownership metrics, and prediction models associated with the instrumental variables approach.

The analysis employs a novel identification strategy that goes beyond unilateral decisions and aggregate shifts in overlapping ownership at the market- or firm-level (Boller and Morton, 2020; Antón et al., 2023; Azar et al., 2025). I first predict ownership structures using a first-order Markovian process on fifteen years of panel data on US shareholdings. I then feed the predictions into the same formulas that yield the overlapping ownership metrics I use throughout the paper, creating analogous versions that serve as instruments.

My empirical strategy distinguishes two margins associated with overlapping ownership. First, the overlap itself, which captures whether managers attend to each other's profits at all. Second, the margin on which managerial attention becomes asymmetric. I present suggestive evidence that the latter dominates, meaning that trading partnerships become more likely when executives weigh more heavily the externalities they impose on the trading partner than the effects on the firm that appointed them in the first place. Furthermore, the channel does not require overlapping shareholders to exert direct influence over executives; it may be a byproduct of their mere presence on both sides of the trading partnership.

I lean on Rotemberg (1984) and insights from the literature on partial vertical integration to argue that decisions to establish or sustain such trading partnerships may enable tunneling opportunities from companies with diffuse ownership to those with concentrated ownership, which can ultimately leave non-overlapping shareholders in those companies worse off. This finding reintroduces the classical tension between ownership and control, shifting the axis of conflict from managers versus owners to overlapping versus non-overlapping shareholders.

Taken together, my findings challenge the interpretation of many results in the existing empirical literature. Prior studies have used composite metrics that conflate the shape and scale margins of overlapping ownership. As a result, many of the mechanisms in those studies may respond to the asymmetric-concentration margin rather than overlapping shareholders per se.

This paper also indicates promising directions for future research. For instance, it remains unclear whether conflicted managers engage in cash-flow tunneling, or whether overlapping shareholders prioritize other goals. The literature on partial vertical integration suggests a few mechanisms worth exploring, e.g., foreclosing companies outside their portfolios, deterring new entrants, or gaining further control over the company, in the supply chain, or across markets.

Furthermore, the empirical exercises cannot pin down the exact channel in play. This gap prompts the question of whether non-overlapping shareholders suffer direct losses or foregone profits—especially because trading partnerships tend to create value for the parties involved. Even if overlapping shareholders capture a significant share of the value generated as private benefits, companies with diffuse ownership may still reap some of the gains.

Finally, it remains to be seen whether the mechanisms identified in the literature apply when trading partners have asymmetric managerial incentives, or whether this interpretation may yield new channels. In particular, mechanisms associated with the disclosure of private information (Boone and White, 2015; Pawliczek et al., 2022), say-on-pay (Fichtner et al., 2017; Bubb and Catan, 2022; Antón et al., 2023), and interlocking boards (Azar, 2022) seem better suited when overlapping shareholders aim for coordination rather than leveraging a one-sided relationship. Other promising avenues include managers on the payroll of institutional funds, selective omission (Wells, 2016; Hemphill and Marcel, 2019), and, especially, tying the compensation of executives to ESG-based outcomes (Bebchuk and Tallarita, 2022; Cohen et al., 2023), which can restrict customers or suppliers to a selected set of companies that meet very specific characteristics.

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APPENDIX

A. Framework: Derivation of profit weights

This appendix depicts the steps to derive profit weights from a framework first introduced by Rotemberg (1984). For a more complete description of the model's assumptions and properties, see Backus et al. (2021).

Shareholders have no particular interest in any single company in their portfolios. From their standpoint, the goal is to maximize cash flow rights across all companies in which they hold shares, which amounts to the overall market value of their portfolios. Hence, let V_{it} represent the total portfolio value of investor $i \in \mathcal{I}$ during year t, which can be written as

$$V_{it}(x_{jt}, x_{-jt}; \beta_t) = \sum_{\forall j \in \mathcal{J}} \beta_{ijt} \pi_{jt}(x_{jt}, x_{-jt}) \tag{A.1}$$

where β_{ijt} designates the fraction of outstanding shares that investor i holds in the company $j \in \mathcal{J}$ in year t. In addition, π_{jt} depicts the profits accrued by company j in year t, which jointly depends on an arbitrary set of managerial decisions, x_{jt} and x_{-jt}. The former represents decisions made by executives appointed by company j, while the latter represents decisions taken by executives appointed in all other companies in the economy.

The author posits that, contrary to what is typically assumed in the literature, shareholders appoint managers not to maximize company profits but to pursue shareholder welfare (Rotemberg, 1984). This claim resonates with several statements in the corporate finance and law literature: that a firm should always answer to its investors (Jensen and Meckling, 1976), and it also anticipates a shift in power dynamics between shareholders and managers that has occurred since the late 1990s. For instance, Wells (2016) lists a series of events that contributed to the rise of shareholder power and the crystallization of two critical instruments for stockholder activism: jawboning and shareholder proposals.[18]

The first began in 1992, when the SEC revised the proxy solicitation rules. While the original goal was to ease communication between large shareholders, they inadvertently promoted the proliferation of meetings and other less formal, less visible interactions between shareholders and corporate management. The second became widely used in the mid-1990s, leading to the formation of several shareholder coalitions that discuss and draft proposals daily.[19]

Thus, the framework proposes a setting in which managers maximize a weighted average of the overall market value of shareholders' portfolios. The objective function of managers builds on the work of Drèze (1974) and Grossman and Hart (1979), so that managers maximize the following expression

$$Q_{jt}\left(x_{jt}, x_{-jt}; \gamma_t\right) = \sum_{\forall i \in \mathcal{I}} \gamma_{ijt} \;\; V_{it}(x_{jt}, x_{-jt}) \tag{A.2}$$

where γ_{ijt} measures the control that investor i exercises in company j in year t. There-

[18] For example, the author mentions the creation of the Institutional Shareholder Services (ISS) in 1985, the issuance of the *Avon Letter* by the US Department of Labor in 1988, the enactment of the Dodd-Frank Act by the US Congress in 2010, and the closer ties between unions and institutional shareholders—fostered by an increasing focus on pension funds since the mid-1990s. For a more in-depth explanation of how these changes affected the corporate power dynamics, see Wells (2016).

[19] For example, James McRitchie, founder of the blog Corporate Governance, acted as the plaintiff in a lawsuit against Meta Platforms Inc. executive officers, including its CEO, Mark Zuckerberg. The accusation is that board members own an excessive shareholding in the company, thereby making decisions that, though beneficial for Meta, ignore their effects on stockholders' portfolios. This is the exact mechanism proposed in Rotemberg (1984). For the full version of the document, refer to this link.

fore, when managers seek to maximize shareholder welfare through their actions, x_{jt}, they use γ_{ijt} to reflect the relative importance of each specific portfolio.

Combining Equations (A.1) and (A.2) yields an expression for the profit weights managers assign to other companies in the economy, which in turn constrain the decisions that managers can take when those decisions affect other firms' outcomes.

$$\begin{aligned} Q_{jt}\left(x_t;\beta_t,\gamma_t\right) &= \sum_{\forall i\in\mathcal{I}} \gamma_{ijt}\ V_{it}(x_t) \\ &= \sum_{\forall i\in\mathcal{I}} \gamma_{ijt} \sum_{\forall k\in\mathcal{J}} \beta_{ikt}\pi_{kt}(x_t) \\ &= \sum_{\forall k\in\mathcal{J}} \sum_{\forall i\in\mathcal{I}} \gamma_{ijt}\ \beta_{ikt}\ \pi_{kt}(x_t) \\ &\propto \pi_{jt}(x_t) + \sum_{k\in\mathcal{J}-\{j\}} \left(\frac{\sum_{\forall i\in\mathcal{I}} \gamma_{ijt}\ \beta_{ikt}}{\sum_{\forall i\in\mathcal{I}} \gamma_{ijt}\ \beta_{ijt}}\right)\pi_{kt}(x_t) \\ &\propto \pi_{jt}(x_t) + \sum_{k\in\mathcal{J}-\{j\}} \kappa_{jkt}\ \pi_{kt}(x_t) \end{aligned}$$

where x_t denotes both the endogenous vector of managerial decisions x_{jt}—for executives appointed to company j—and the corresponding set of response functions for companies in $\mathcal{J}-\{j\}$: x_{-jt}.[20] The interpretation of κ_{jkt} is straightforward. Profit weights determine how the manager appointed by company j weights the profits of company k in year t when making an arbitrary set of decisions x_{jt} that impose externalities on that company's profits.

Furthermore, using vector notation, I can rewrite the expression for profit weights as a ratio of two inner products. Interestingly, this transformation into polar notation yields a decomposition that clearly illustrates the underlying mechanisms.

$$\begin{aligned} \kappa_{jkt} &= \frac{\sum_{\forall i\in\mathcal{I}} \gamma_{ijt}\ \beta_{ikt}}{\sum_{\forall i\in\mathcal{I}} \gamma_{ijt}\ \beta_{ijt}} \\ &= \frac{\langle \gamma_{jt}, \beta_{kt}\rangle}{\langle \gamma_{jt}, \beta_{jt}\rangle} \\ &= \frac{cos(\gamma_{jt},\beta_{kt})\ \cancel{\|\gamma_{jt}\|}\ \|\beta_{kt}\|}{cos(\gamma_{jt},\beta_{jt})\ \cancel{\|\gamma_{jt}\|}\ \|\beta_{jt}\|} \end{aligned}$$

The first factor is a shape-only statistic that compares the overlap between control and cash flow rights across companies. The denominator tells the manager in company j how similar control and cash flow rights are within the same firm that appointed them. The numerator, instead, uncovers the overlap between control rights in company j and cash flow rights in company k. Therefore, a higher magnitude on the first factor indicates that managers at both company j and company k pay attention to each other's profits. The second factor is a scale-only statistic that represents ownership concentration (in terms of cash flow rights) across the two companies. This object maps exactly to RSC in Equation (1). So, a higher magnitude on the second factor has two implications. First, the manager in company j pays closer attention to how their decisions affect company k. Second, company k becomes more self-centered, and its manager pays less attention to the effects on company j. Therefore, this metric

[20] The fourth step in the derivation stems from a simple normalization, so that every manager assigns a profit weight equal to 1 to the company that appointed them.

captures not only the intensity but also the direction of the attention in managerial decisions.

Note that using profit weights requires appropriately distinguishing between control and cash flow rights. In this paper, I assume proportional control, implying that γ_{jt} equals β_{jt}.[21]

$$\kappa_{jkt} = cos(\beta_{jt}, \beta_{kt}) \frac{\|\beta_{kt}\|}{\|\beta_{jt}\|}$$

As a result, $cos(\gamma_{jt}, \beta_{jt})$ becomes one, and the first factor simplifies to $cos(\beta_{jt}, \beta_{kt})$, which maps exactly to OSS in Equation (1). The final expression for profit weights can be written as

B. Mechanisms associated with partial vertical integration

This appendix connects empirical findings on vertically overlapping ownership with features and mechanisms associated with partial vertical integration. Unlike the paper, which focuses on a single mechanism in which overlapping shareholders extract private benefits along the supply chain, this appendix ties together empirical results across different strands of the literature. I begin by discussing how overlapping shareholders and partially integrated companies can play an analogous role in the supply chain. Then, I present empirical exercises inspired by prior evidence, in which overlapping shareholders—like partially integrated firms—can help trading partnerships by reducing contractual and informational frictions that typically arise in vertical relationships, such as holdup and double marginalization.

The industrial organization literature shows that vertical mergers can affect upstream and downstream market outcomes (Bolton and Whinston, 1993; Boehm and Sonntag, 2023). Likewise, partial vertical integration, through partial equity stakes rather than full acquisitions, and vertically overlapping ownership, through third-party shareholders, can produce similar effects.

A good example of how vertically overlapping ownership and partial vertical integration share common features and can yield similar outcomes materializes in the work of Fee et al. (2006) and Freeman (2025). The former shows that partially integrated customers and suppliers form trading partnerships with longer lifespans, without resorting to mergers and acquisitions. The latter extends the analysis to third-party overlapping shareholders. Needless to say, the analogy can be extrapolated to other mechanisms associated with typical features of vertical integration and vertical control.[22]

To begin with, overlapping shareholders can enable valuable trading partnerships that would not otherwise exist because of contractual frictions, such as incomplete contracts or injunctions from competition policy authorities.[23] For example, overlapping shareholders can ease access to private corporate data, reducing the cost of getting information about prospective trading partners or alleviating conflicting interests; facilitate screening processes and reduce transaction costs, making it more likely for companies

[21] The assumption is not harmless, but it has been standard in the literature for many years. A few papers, however, have pursued a different direction. For example, Gilje et al. (2020) derives an alternative metric that incorporates attentiveness through a corporate voting model, while Antón et al. (2025*b*) represents control rights as a quadratic form of cash flow rights. For a discussion of the tradeoffs between cash flow and control rights—incentives versus influence in tunneling—see Newham et al. (2025).

[22] Vertical integration is a way of organizing a trading relationship in which a company aims to gain control of multiple steps along the supply chain. Vertical control, by contrast, involves transferring decision-making rights over some, but not all, aspects of the trading partnership.

[23] As an example, revenue-sharing contracts have been under scrutiny by competition policy authorities in both the US and Europe. The Directorate-General for Competition of the European Commission accused Google in 2018 of using these contracts to pay device manufacturers, on the condition that they pre-install Google Search across their portfolios and do not install competing search engines (Case AT.40099, Google Android.)

to select each other; and substitute contracts, mergers, and acquisitions that are too costly or competition policy authorities would deem as anticompetitive.

Overlapping shareholders can also induce companies to reduce their exposure to systemic supply-chain risk. Managers can then learn the benefits of redundant trading partners in reducing the impact of natural disasters (Carvalho et al., 2021; Barrot and Sauvagnat, 2016), bullwhip effects (Croson and Donohue, 2005), sales dependence, or other external shocks (Wang, 2025). Executives can also enhance sourcing practices, such as increasing supply reliability (Bolton and Whinston, 1993; Wu and Birge, 2014), easing cash constraints (Fee et al., 2006), or securing access to essential facilities or networks.

Beyond these positive effects, Subsection III.B discusses typical drawbacks of partial vertical integration. For instance, foreclosure in upstream and downstream markets (Levy et al., 2018; Boehm and Sonntag, 2023) or tighter market-entry barriers (Newham et al., 2025). Throughout the paper, I denote these negative effects as tunneling, even though they need not involve cash-flow transfers. I leave the discussion and analysis of such mechanisms to that section.

In the remainder of the appendix, I focus on two mechanisms—holdup and double marginalization—and present suggestive evidence consistent with prior research. I use this comparison to set a benchmark and show that my empirical approach aligns with previous findings, particularly from studies that leverage better settings and sources of variation for these specific mechanisms. Nevertheless, the analysis also indicates that neither holdup nor double marginalization fully explains the causal relationship between ownership structures and trading partnerships, implying that all mechanisms potentially coexist.

B.1. Holdup

Customers and suppliers can often boost profits through coordination and long-term contracting. However, this may not always be possible, as contractual frictions often preclude companies from pursuing these opportunities. In particular, information asymmetries often make it difficult for them to agree on the contract's content or who should retain residual control.

When long-term contracts are unavailable, mergers and acquisitions can offer a workaround, at least when the expected benefits outweigh the associated costs. Even then, competition policy authorities can freeze proceedings because of anticompetitive concerns.

Holdup constitutes a typical example of contractual frictions between customers and suppliers. When both parties benefit from the agreement but only one bears the costs, managers cannot credibly commit without a binding document. Otherwise, sooner or later disagreements over how to split the gains will arise, leading to renegotiations that often harm the company making the partnership-specific investment. This situation gives rise to the name of this contractual friction, as one company fears being held up by the other party.

Previous research has found evidence that overlapping shareholders limit holdup by reducing information asymmetries. For example, Freeman (2025) shows that overlapping shareholders lengthen trading partnership lifespans while improving innovation outcomes, while Chen (2024) and Deng et al. (2025) find that suppliers invest more intensively in partnership-specific assets.

Although the exact channel for easing information asymmetries remains unclear, overlapping shareholders can presumably resort to activities documented in other studies: disclosing otherwise private information (Pawliczek et al., 2022; Boone and White, 2015) or enforcing tacit agreements by other means, such as voting (Fichtner et al.,

2017; Bubb and Catan, 2022; Antón et al., 2023), interlocking boards (Azar, 2022), tying compensation to ESG-based outcomes (Bebchuk and Tallarita, 2022; Cohen et al., 2023), or through selective omission (Wells, 2016; Hemphill and Marcel, 2019).[24]

Next, I examine whether overlapping shareholders increase the likelihood of trading partnerships in settings affected by holdup. To do so, I estimate and compare heterogeneous fostering effects of a higher OSS across different subsamples where holdup is more or less prevalent.

Before moving forward, notice that the literature recognizes three dimensions of innovation activities associated with holdup (Aghion et al., 2013; Chen, 2024; Freeman, 2025; Deng et al., 2025; Antón et al., 2025*a*). *Innovation input* reflects the resources and effort firms invest; *innovation output* represents the outcomes of the innovation process; and *innovation specificity* refers to the degree of customization a trading partner requires.

For this empirical exercise, I measure the prevalence of holdup by classifying companies by their R&D intensity. Specifically, I use the ratio of annual R&D expenses to total book assets, a standard proxy for innovation input in the literature. However, using only innovation input has a caveat. This metric captures only the average likelihood that a company will face holdup with any given trading partner, unlike input specificity, which provides pairwise measures.

Therefore, I cannot classify customer-supplier dyads by holdup risk and must instead rely on customers' and suppliers' innovation inputs. To avoid dyads ending in more than one subsample, I fix each company's R&D intensity in the first year I observe a non-missing value and keep only years after that first observation. If a company has no non-missing value between 1999 and 2012, I impute the median value within 4-digit SIC industries. Then, to avoid outliers, I classify customers and suppliers as above or below the average R&D intensity among the remaining firms in the sample, excluding the top decile (8.30%). Finally, I split the sample described in Section I, assigning each dyad sc to one of four possible subsamples $g(sc) \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\}$, based on all possible combinations (customer/supplier and above/below average R&D intensity).

I estimate an analogous version of Equation (2), separately on each subsample, using OLS and 2SLS across the three fixed-effects specifications detailed in Subsection II.A,

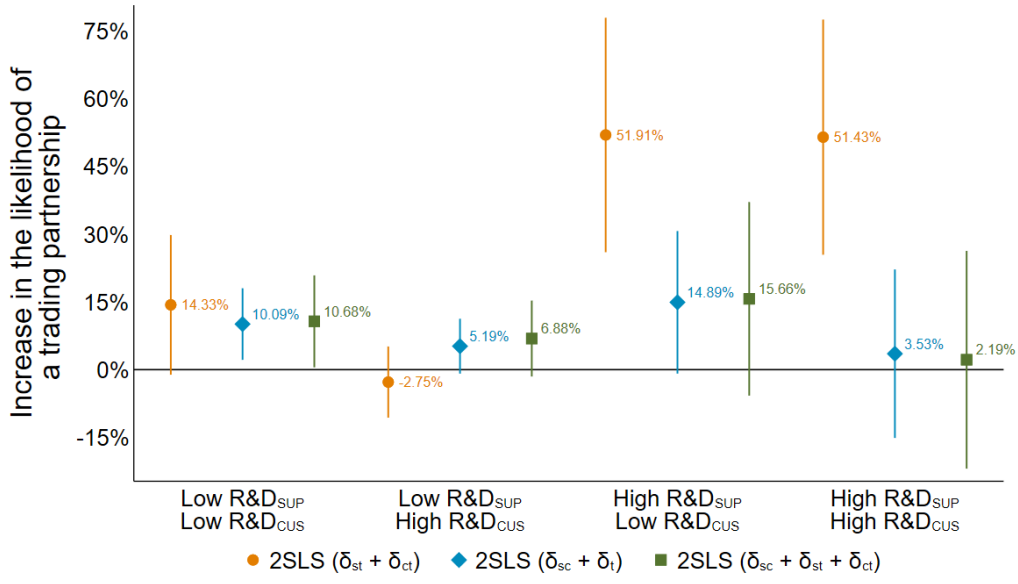
$$\text{LTP}_{sct}^{g(sc)} \;=\; \delta_{FE}^{g} \;+\; \tau^{g} \cdot \text{OSS}_{sct}^{g(sc)} \;+\; \varepsilon_{sct}^{g(sc)}, \quad g \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\} \tag{B.1}$$

so that τ^g reports a one-standard-deviation increase in OSS on the pooled average likelihood of a trading partnership. For details, see Table SA.4 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages.

Figure B.1 shows that, for my preferred specification (orange), an increase in OSS has a larger impact when suppliers have an R&D investment above the average, consistent with the results of Chen (2024), Freeman (2025), and Deng et al. (2025), who find that overlapping shareholders facilitate innovation only when investments hinge on the supplier side. In contrast, specifications that include dyad fixed effects (blue and green) report non-statistically significant effects, which may be due to the lack of within-dyad variation in these two subsamples.

[24] *Interlocking boards* refers to institutional shareholders who have a seat on the board of multiple firms. *ESG-based compensation* ties executive pay to environmental, social, and governance metrics, which can restrict customers or suppliers to companies meeting very specific characteristics. *Selective omission* represents situations in which shareholders encourage actions that simultaneously benefit the company's profits and investors' portfolios, but remain silent otherwise. Wells (2016) provides a few examples that illustrate how institutional investment funds tipped the scales and amplified their voice in corporate decision-making through several means at their disposal, e.g., jawboning, shareholders' proposals, proxy voting, among others.

Figure B.1 – *Vertically overlapping ownership alleviates holdup*



Notes: The figure depicts the effect of a one-standard-deviation increase in *ownership-structure similarity* (OSS) on the average likelihood of a trading partnership across four settings representing different degrees of holdup. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I fix each company's R&D intensity in the first year I observe a non-missing value and keep only years after that first observation. If a company has no non-missing value between 1999 and 2012, I impute the median value within 4-digit SIC industries. I classify customers and suppliers above or below the average R&D intensity among the remaining firms in the sample, excluding the top decile (8.30%), and then split the sample, assigning each dyad *sc* to one of four possible subsamples $g(sc) \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\}$. I define OSS in Equation (1). I estimate Equation (B.1) separately on each subsample, using OLS (left, gray) and 2SLS (right, color) across three FE specifications, as described in Subsection II.A: (customer×year)-FE and (supplier×year)-FE (orange circle), dyad-FE and year-FE (blue diamond), and both combined (green square). The 2SLS instrument is an analogous version of OSS, based on predicted ownership structures, as described in Subsection II.C. Confidence intervals report dyad-cluster robust standard errors at the 95% confidence level. All regressions drop singleton observations in each subsample under the most stringent FE specification (green square). For further reference, see Subsection B.1 for a description of the analysis and Table SA.4 in the Appendix for details.

More importantly, coefficients are positive and statistically significant (the orange specification, controlling for time-varying company-FE, has a p-value of 0.069) when both customers and suppliers have a below-average R&D intensity. A one-standard-deviation increase in OSS increases the pooled average likelihood of a trading partnership by around 10.09% to 14.33%, hinting that holdup cannot fully explain the role of overlapping shareholders.

B.2. Double marginalization

When customers and suppliers operate in less competitive markets, they apply successive markups to their marginal costs, leading to a problem known as *double marginalization*. Although not a general rule, double marginalization tends to reduce profits for all companies along the supply chain (Hamilton and Mqasqas, 1996). Thus, firms seek to avoid double marginalization through different means, such as downstream-profit revenue-sharing contracts and nonlinear pricing.[25]

[25] Revenue-sharing contracts appeal to companies because they can incorporate clauses to address the specificities associated with trading partnerships, such as product quality, technical specifications, or retail services. Easing double marginalization is another benefit of this type of contract.

However, when competition authorities do not outright forbid them, these contracts are usually hard to enforce. The core issue is that customers and suppliers need full information about each other and their actions, which becomes increasingly complex as the supply chain grows longer. Therefore, like holdup problems, double marginalization reflects contractual friction that overlapping shareholders can help overcome.

For this empirical exercise, I retrace the holdup methodology, using market concentration as an approximate indicator of competition. My strategy deviates from the traditional approach, which requires a prescriptive market definition. Instead, I use a version of the Herfindahl-Hirschman Index (HHI) of market concentration proposed by Hoberg and Phillips (2016). By using product descriptions in 10-K filings, the authors build product-differentiation distances, which in turn define the degree of competition between any two publicly listed US companies. Consequently, every company in the sample can have competitors across different industries, depending on the goods and services it produces.

Market concentration is then defined at the firm level by calculating the HHI across pairwise measures of product similarity (HHI-TNIC), which capture the degree of competition each company faces across industries and products. For comparability, the authors exclude close-competitor dyads with low similarity in the product space, so that the coarseness of their data matches that of 3-digit SIC code industries.

A prominent feature of the approach in Hoberg and Phillips (2016) is that the network definition makes it unnecessary to specify juxtaposed markets across industries. This is most appropriate for this setting, where publicly listed US companies are multi-product firms with few close competitors in each market. This implies that assigning a single market to each company can introduce significant errors. First, it underestimates the number of close competitors and replaces it with potentially many companies producing unrelated goods and services. Second, it overestimates concentration for companies that produce niche goods and services in industries with many other large firms. For more details on text-based product similarity and its properties, see Hoberg and Phillips (2016), Frésard et al. (2020), and Ederer and Pellegrino (2025).

To avoid dyads ending in more than one subsample, I fix each company's HHI-TNIC in the first year I observe a non-missing value and keep only years after that first observation. If a company has no non-missing value between 1999 and 2012, I drop all observations for that firm. Then, I classify customers and suppliers as above or below 1,500 HHI-TNIC, which corresponds to the threshold that separates *unconcentrated* and *moderately concentrated* markets, according to the Horizontal Merger Guidelines, issued by the US Department of Justice (DOJ) and the Federal Trade Commission (FTC) in 2010. Finally, I split the sample described in Section I, assigning each dyad sc to one of four possible subsamples $h(sc) \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\}$, based on all possible combinations (customer/supplier and above/below 1,500 HHI-TNIC).

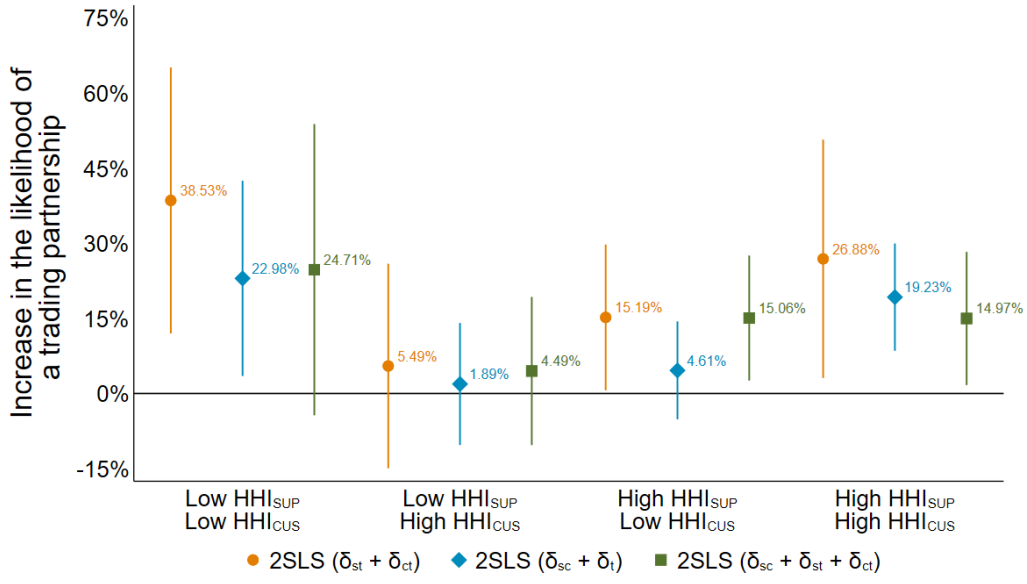
I estimate an analogous version of Equation (2), separately on each subsample, using OLS and 2SLS across the three fixed-effects specifications detailed in Subsection II.A,

$$\text{LTP}_{sct}^{h(sc)} \;=\; \delta_{FE}^{h} \;+\; \tau^{h} \cdot \text{OSS}_{sct}^{h(sc)} \;+\; \varepsilon_{sct}^{h(sc)}, \quad h \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\} \tag{B.2}$$

so that τ^h reports a one-standard-deviation increase in OSS on the pooled average likelihood of a trading partnership. For details, see Table SA.5 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages.

Figure B.2 suggests that overlapping shareholders play an important role when downstream and upstream markets are moderately or highly concentrated, consistent with the premise that they can help offset losses from double marginalization (Riva, 2022).

Figure B.2 – *Vertically overlapping ownership alleviates double marginalization*



Notes: The figure depicts the effect of a one-standard-deviation increase in *ownership-structure similarity* (OSS) on the average likelihood of a trading partnership across four settings representing different degrees of double marginalization. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I fix each company's HHI-TNIC in the first year I observe a non-missing value and keep only years after that first observation. I classify customers and suppliers above or below 1,500 HHI-TNIC, which corresponds to the threshold that separates *unconcentrated* and *moderately concentrated* markets, and then split the sample, assigning each dyad sc to one of four possible subsamples $h(sc) \in \{\text{LL, LH, HL, HH}\}$. I define OSS in Equation (1). I estimate Equation (B.2) separately on each subsample, using OLS (left, gray) and 2SLS (right, color) across three FE specifications, as described in Subsection II.A: (customer×year)-FE and (supplier×year)-FE (orange circle), dyad-FE and year-FE (blue diamond), and both combined (green square). The 2SLS instrument is an analogous version of OSS, based on predicted ownership structures, as described in Subsection II.C. Confidence intervals report dyad-cluster robust standard errors at the 95% confidence level. All regressions drop singleton observations in each subsample under the most stringent FE specification (green square) or when a company has no non-missing HHI-TNIC between 1999 and 2012. For further reference, see Subsection B.2 for a description of the analysis and Table SA.5 in the Appendix for details.

Estimates show that a one-standard-deviation increase in OSS raises the average likelihood of a trading partnership by 14.97% to 26.88%, depending on the fixed-effects specification.

Nevertheless, the empirical exercise offers no support for the hypothesis that double marginalization fully explains the role of overlapping shareholders in vertical relations. When companies operate in unconcentrated upstream and downstream markets, point estimates are positive and statistically significant (the green specification, controlling for dyad-FE and time-varying company-FE, has a p-value of 0.096), with coefficients ranging between 22.98% and 38.53%. Similarly, two of three specifications indicate that overlapping shareholders also make trading partnerships more likely when the downstream market is unconcentrated, but the upstream market is moderately or highly concentrated. Coefficients for my preferred specification (orange) and the fully saturated specification (green) are roughly 15%, though the dyad-FE specification (blue) reports a non-statistically significant coefficient.

Taken together, my results support previous findings that overlapping shareholders can help customers and suppliers overcome informational and contractual difficulties that usually require more cumbersome arrangements. As mentioned above, neither holdup nor double marginalization can fully explain the fostering results in Subsection

III.A, so my findings do not rule out the possibility that overlapping shareholders can play a significant role in other margins—and that all mechanisms can occur simultaneously. This includes the evidence in Subsection III.B that overlapping shareholders can leverage asymmetric managerial incentives, as well as other contractual and informational frictions or risk exposure along the supply chain. If anything, the evidence points to the importance of further exploring additional channels.

C. Robustness

This appendix collects the robustness checks that support the paper's main findings. The first three subsections cover methodological decisions described in the paper: alternative sample definitions, overlapping-ownership metrics, and prediction models for the instrument. The last subsection describes a robustness exercise that drops companies involved in or related to additions and deletions of the S&P 500 index. Throughout, the paper's qualitative conclusions remain unchanged.

C.1. Alternative sample definitions

This subsection presents four alternative criteria for defining the sample, focusing on comparing fostering effects. The first two sample definitions arise from discrepancies in two open data sources, which I use to identify feasible customer-supplier dyads and trading partnerships. By contrast, the next two sample definitions stem from an alternative construction based on the historical supply chain structure and a text-based network definition of close competitors.

Feasible-dyads and fully-weighted-partnerships samples. After merging the two available open data sources on vertical upstream relatedness (Frésard et al., 2020) and the Customer Segment (Barrot and Sauvagnat, 2016), a small fraction of trading partnerships show null vertical upstream relatedness. On the one hand, the first study mentions that the procedure loses track of a small fraction of dyads when mapping CUSIP (American Bankers Association) to GVKEY (Compustat) identifiers. On the other hand, the second publication explains that using phonetic string-matching algorithms can lead to incongruent trading partnerships, even after manual review. Thus, discrepancies arising when merging the data may reflect errors in one or both sources.

Throughout the main analysis, I have established a compromise solution. First, I define all trading partnerships as feasible by default. Second, for customer-supplier dyads with a null vertical upstream relatedness, I impute them the average value among trading partnerships operating in the same year. To assess how determinant this assumption is for my findings, I next consider two distinct reweights of the sample that align with scenarios in which errors arise solely from one data source.

Should inconsistencies arise from improper mapping between companies' names and GVKEY identifiers, I define the *Feasible dyads sample* by excluding all infeasible customer-supplier dyads, even if they appear in the trading partnerships dataset. Instead, if discrepancies arise in mapping vertical upstream relatedness between CUSIP and GVKEY identifiers, I define the *Fully-weighted partnerships sample*, in which I assign an observational weight equal to one to all trading partnerships in the sample.

Historical supply chain sample. For this sample, rather than using observational weights, I identify a subset of customer-supplier dyads most likely to be feasible given the historical structure of the US supply chain. I begin by blending all trading partnerships between 1976 and 2016 into a unique supply chain network—a common technique in the literature on *link analysis*. Then, I proceed as follows. Consider any arbitrary customer c. I retrieve the set of historical suppliers, $\mathcal{S}(c)$, from the full network. Then, for every company $j \in \mathcal{S}(c)$, I retrieve its set of historical customers, $\mathcal{C}(j)$. Thus, the

union $\bigcup_{j \in S(c)} \mathcal{C}(j)$ defines a set of companies closely related to c, irrespective of the industry in which they operate. Finally, for every company $k \in \{\mathcal{C}(j) : j \in \mathcal{S}(c)\}$, I retrieve its set of historical suppliers, $\mathcal{S}(k)$. This last step yields the following set of companies

$$\left\{ \bigcup_{j \in S(c)} \bigcup_{k \in S(j)} \mathcal{S}(k) \right\}$$

which I define as the set of feasible suppliers of c.

After repeating the procedure for every customer in the Customer Segment, I retain only annual observations between 1999 and 2013, in which both companies simultaneously report book assets to the SEC, in industries other than the financial, transportation, communication, and utility sectors.

TNIC-based competitors sample. For this sample, I identify feasible trading partners based on their close competitors, without employing observational weights. I begin by constructing year-specific sets of close competitors, $\mathcal{P}(jt)$, for every company j in the Customer Segment, excluding those in the financial, transportation, communication, and utility industries. Using text-based product similarity metrics from Hoberg and Phillips (2016), I define company k as belonging to $\mathcal{P}(jt)$ if j and k are closely related in the product space in year t. That is, if the pairwise product similarity between the two companies is strictly positive.

Next, I proceed in two steps. First, for any given year t and customer c with an observed trading partnership, I define that every supplier of c, $\mathcal{S}(ct)$, is a feasible supplier of every close competitor of c, $\mathcal{P}(ct)$. Next, I proceed analogously for every supplier in the Customer Segment.

Thus, for every year t, the union

$$\left\{ \bigcup_{\forall\, c} \mathcal{P}(ct) \times \mathcal{S}(ct) \right\} \quad \bigcup \quad \left\{ \bigcup_{\forall\, s} \mathcal{C}(st) \times \mathcal{P}(st) \right\}$$

represents the subset of yearly feasible customer-supplier dyads in this sample.

Robustness. I estimate Equation (2) across all five samples—including the one from the main analysis—and the three fixed-effects specifications detailed in Subsection II.A, using OLS and 2SLS. For details, see Table SA.6 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages.

Figure C.1 depicts coefficients across all regressions, showing that conclusions remain qualitatively unchanged. Quantitatively, however, two samples consistently display smaller effects and have non-statistically significant point estimates in specifications that include dyad fixed effects.

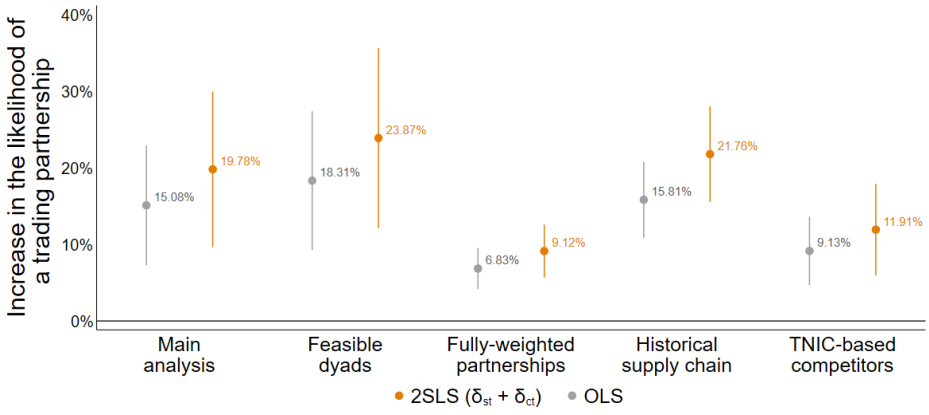
In part, the smaller effects in the *TNIC-based competitors* and *fully-weighted partnerships* samples may be explained by an average probability of a trading partnership that is 85 to 116 times higher than in the main analysis, making the comparison—even across relative effects—less straightforward.

In the *TNIC-based competitors* sample, estimates are significant only at the 90% confidence level, with a p-value of 0.075 in the dyad-FE specification (blue) and 0.104 in the fully saturated model (green). However, these results do not seem to contradict the conclusions from the main analysis.

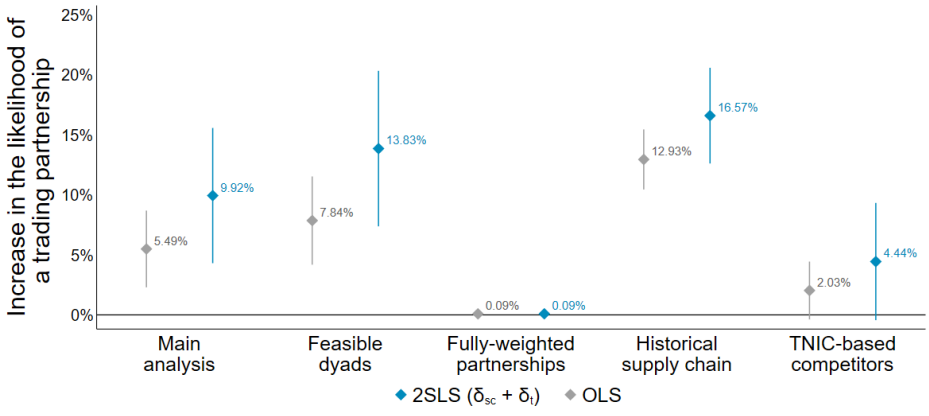
For the *fully-weighted partnerships* sample, estimates are statistically and numerically indistinguishable from zero. A caveat with this sample definition is that customer-supplier dyads that become trading partnerships change not only the outcome, but

Figure C.1 – *Alternative sample definitions do not overturn the effect of ownership-structure similarity (OSS)*

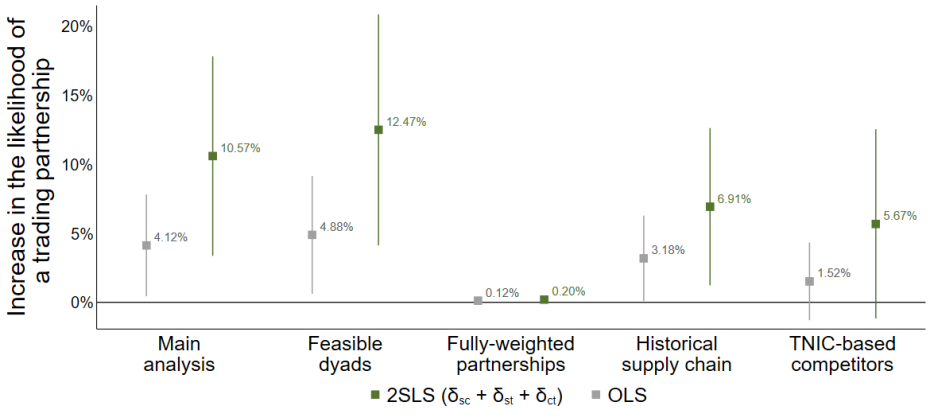
(a) (customer×year)-FE and (supplier×year)-FE



(b) dyad-FE and year-FE



(c) dyad-FE, (customer×year)-FE, and (supplier×year)-FE



Notes: The figure depicts the effect of a one-standard-deviation increase in OSS on the average likelihood of a trading partnership across five sample definitions, based on feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I define OSS in Equation (1). I estimate Equation (2) on each sample, using OLS (left, gray) and 2SLS (right, color) across three FE specifications (one for each panel), as described in Subsection II.A. The 2SLS instrument is an analogous version of OSS, based on predicted ownership structures, as described in Subsection II.C. Confidence intervals report dyad-cluster robust standard errors at the 95% confidence level. All regressions drop singleton observations in each sample under the most stringent FE specification (green square). For further reference, see Subsection C.1 for a description of the analysis and Table SA.6 in the Appendix for details.

also their presumed probability of being feasible, $\pi(F_{sct} = 1)$.[26] Therefore, dyad-FE absorb the feasibility probability during estimation, killing all within-dyad variation in the outcome in the process.

Taken together, and regardless of quantitative disparities, this subsection points to the same qualitative conclusion: overlapping shareholders increase the likelihood of trading partners among companies in their portfolios.

C.2. Alternative overlapping ownership metrics

For the main analysis, I introduce *ownership-structure similarity* (OSS), a metric based on the profit-weight derivation from the framework in Rotemberg (1984). In this subsection, I test my conclusions using other standard measures in the literature: overlapping market value and overlapping-shares product. Since these metrics focus on cross-company overlap in ownership structures, I limit the comparison to the fostering-effects analysis.

Overlapping market value (OMV). Antón and Polk (2014) and Freeman (2025) use this metric, which represents the fraction of the total market value of supplier s and customer c held by overlapping shareholders, $\mathcal{O} \subset \mathcal{I}$. Namely,

$$\text{OMV}_{sct} \;=\; \frac{\sum_{i \in \mathcal{I}} \left(V_{ist} \;+\; V_{ict}\right) \{i \in \mathcal{O}\}}{V_{st} \;+\; V_{ct}}$$

where V_{jt} denotes the total market value of company j in year t.

Overlapping-shares cross-product (OSCP). Hansen and Lott (1996) and Freeman (2025) employ this metric, which quantifies the cross-product of customer and supplier ownership structures. That is,

$$\text{OSCP}_{sct} \;=\; \sum_{i \in \mathcal{I}} \beta_{ist} \times \sum_{i \in \mathcal{I}} \beta_{ict}$$

where β_{ijt} designates the fraction of outstanding shares that investor $i \in \mathcal{I}$ holds in the company $j \in \mathcal{J}$ in year t.

Robustness. I estimate Equation (2) across all three regressors—including OSS from the main analysis—and the three fixed-effects specifications detailed in Subsection II.A, using OLS and 2SLS. For details, see Table SA.7 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages.

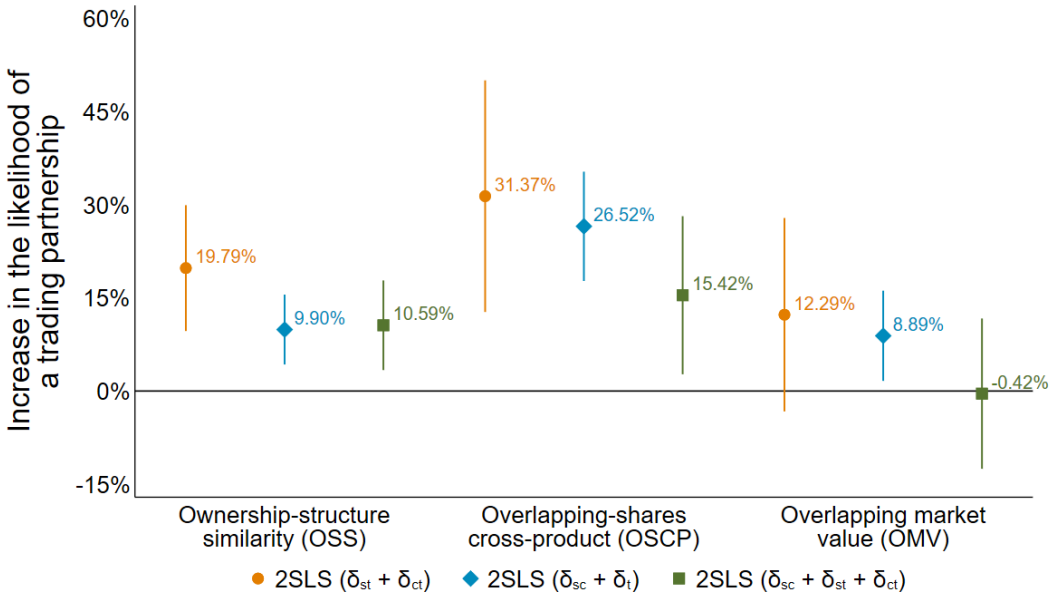
Figure C.2 displays coefficients consistent with the findings in Freeman (2025), while also highlighting the merits of OSS in capturing ownership overlap. To begin with, while closely related to profit weights, OSCP conflates OSS and RSC into a single metric.[27] It comes to no surprise then that OSCP reports point estimates that are roughly 50% larger than those of OSS. In addition, my identification strategy does not address potential simultaneity between trading partnerships and market capitalization, so OMV estimates may suffer from measurement error and endogeneity that systematically bias these coefficients toward zero.

In any case, these results do not appear to contradict the conclusions from the main analysis.

[26] For a proper definition of $\pi(F_{sct} = 1)$, see Subsection II.B.

[27] OSS can be written as the cross-product of customer and supplier ownership structures divided by the norm of each ownership structure.

Figure C.2 – *Alternative overlapping-ownership metrics do not overturn the effect of ownership-structure similarity (OSS)*



Notes: The figure depicts the effect of a one-standard-deviation increase in ownership overlap on the average likelihood of a trading partnership across three definitions of overlapping ownership. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I define OSS in Equation (1); OSCP and OMV in Subsection C.2. I estimate Equation (2) using 2SLS across three FE specifications, as described in Subsection II.A: (customer×year)-FE and (supplier×year)-FE (orange circle), dyad-FE and year-FE (blue diamond), and both combined (green square). The 2SLS instruments are analogous versions of each metric, based on predicted ownership structures, as described in Subsection II.C. Confidence intervals report dyad-cluster robust standard errors at the 95% confidence level. All regressions drop singleton observations under the most stringent FE specification (green square). For further reference, see Subsection C.2 for a description of the analysis and Table SA.7 in the Appendix for details.

C.3. Alternative prediction models for the instrument

A potential threat to identification is that the instrument may be sensitive to the empirical model used to predict shareholdings. The exclusion restriction assumes that supply chain persistence decays faster than ownership persistence. Therefore, if using only one lag were not enough, my instrumental variables approach would not address endogeneity properly. Another minor concern relates to misspecification. In particular, if the model were not flexible enough, I would be excluding much of the non-endogenous variation in ownership from the instrument.

In this subsection, I test these two hypotheses on the fostering effects, comparing those results against versions using alternative empirical models for the prediction exercise. Specifically, I replicate the analysis using instruments derived from a single-coefficient second-order Markovian process and from a first-order Markovian process with investor-specific parameters.

Single-coefficient second-order Markovian process. Two reasons justify testing a second-order Markovian process. First, the instrument captures longer-memory portfolio adjustments, making it more likely to satisfy the exclusion restriction assumption. If the new results do not flip the sign or significance of the 2SLS estimates, it would lend credibility to the claim that the main analysis satisfies the exclusion restriction too. Second, because both instruments handle measurement error, changes in point estimates can potentially reveal more about the direction and nature of the bias.

An important caveat, however, is that the sample loses an additional year, so that this approach is best read as a robustness check rather than the baseline strategy. Nonetheless, the additional lag strengthens the interpretation that compliers of this instrument pertain to portfolios of shareholders with more stable holdings over time.

In this case, I use the following empirical model for predicting ownership structures

$$\beta_{ijt} \;=\; \alpha \;+\; \rho \cdot \beta_{ijt-2} \;+\; \nu_{ijt} \tag{C.1}$$

where β_{ijt} represents the fraction of outstanding shares owned by investor $i \in \mathcal{I}$ on company $j \in \mathcal{J}$ in year t. I estimate this equation using OLS on the same ownership network described in Subsection II.C, which yields the following coefficients

$$\underset{}{\alpha} \quad \underset{(0.000012)}{0.0009} \qquad \underset{}{\rho} \quad \underset{(0.002866)}{0.8573} \qquad \underset{}{R^2} \quad 0.729$$

First-order Markovian process with investor-specific parameters. Allowing parameters in a first-order Markovian process to vary across investors complements the previous robustness, alleviating misspecification concerns in three ways. First, it relaxes the assumption that all investors share a common persistence parameter. Thus, estimates reflect heterogeneous portfolio-adjustment speeds across institutions with different mandates, horizons, and rebalancing rules. Second, it provides a soft test of the exclusion restriction assumption. If 2SLS estimates do not change when ownership persistence varies by investor, the identifying assumption—that ownership persistence decays more slowly than that of the supply chain—does not depend on an overly restrictive homogeneity assumption. Third, the single-parameter version could leave systematic, predictable components of shareholding in the residual. This empirical model, however, should predict more of that idiosyncratic variation.

In this case, I use the following empirical model for predicting ownership structures

$$\beta_{ijt} \;=\; \delta_i \;+\; \sum_{k \in \mathcal{I}} \rho_k \cdot \beta_{ijt-1}\big\{i = k\big\} \;+\; \nu_{ijt} \tag{C.2}$$

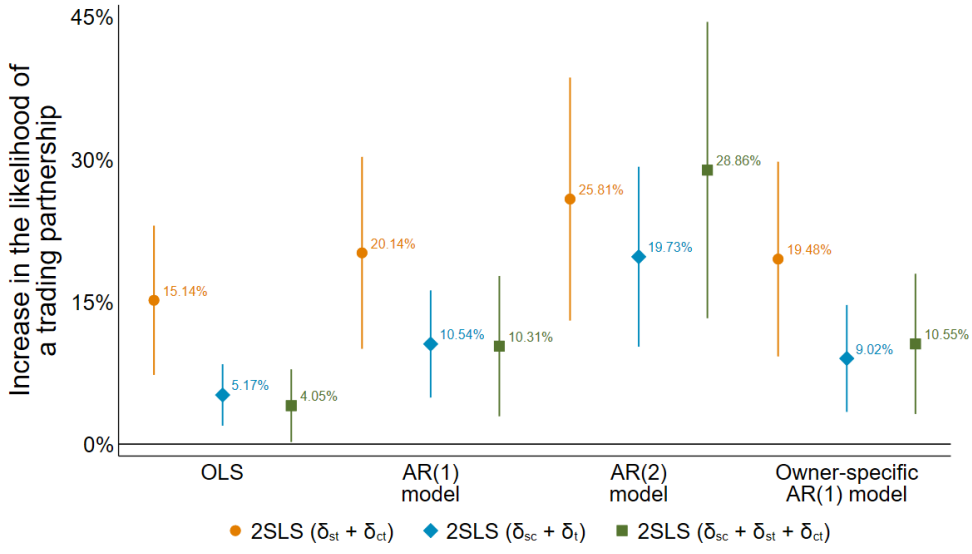
where δ_i represents investor fixed effects. As in the previous specification, I use OLS to estimate parameters for 13,698 distinct investors (the R^2 is 0.864, slightly higher than for the instrument used in the main analysis).

Robustness. Retracing my strategy for the main analysis, I feed the predicted ownership structures back into the formulas described in Subsection I.C. I restrict observations to 2001–2013 to compare results across instruments in a fixed sample. Then, I estimate Equation (2) using OLS and 2SLS—one for each instrument—across the three fixed-effects specifications detailed in Subsection II.A. For details, see Table SA.8 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages.

Figure C.3 implies that the main analysis conclusions do not depend on the prediction model used to construct the instrument. All 2SLS coefficients have the same sign and report similar magnitudes across all three fixed-effects specifications. Nevertheless, the figures display two patterns worth mentioning.

First, using either of the two first-order Markovian processes yields statistically indistinguishable coefficients. This suggests that the exclusion restriction assumption does not depend on persistence being homogeneous across investors. Moreover, it also indicates that the instrument from the main analysis does not leave systematic, predictable shareholding variation in the residual.

Figure C.3 – *Alternative prediction models for the instrument do not overturn the effect of ownership-structure similarity (OSS)*



Notes: The figure depicts the effect of a one-standard-deviation increase in OSS on the average likelihood of a trading partnership using three different instruments. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2001 and 2013, as described in Subsection I.B. I define OSS in Equation (1). I estimate Equation (2) using 2SLS across three FE specifications, as described in Subsection II.A: (customer×year)-FE and (supplier×year)-FE (orange circle), dyad-FE and year-FE (blue diamond), and both combined (green square). The 2SLS instruments are analogous versions of OSS, based on predicted ownership structures, each obtained through a different empirical model, as described in Subsections II.C and C.3. Confidence intervals report dyad-cluster robust standard errors at the 95% confidence level. All regressions drop singleton observations under the most stringent FE specification (green square). For further reference, see Subsection C.3 for a description of the analysis and Table SA.8 in the Appendix for details.

In addition, the second-order Markovian process yields point estimates slightly higher than those of the main analysis. Only the coefficients for the fully saturated specification (green) differ statistically from one another (p-value = 0.035, assuming independence between estimates). Moreover, as both instruments typically account for measurement error, this pattern suggests that OLS estimates are likely downward-biased.

Taken together, these empirical patterns imply that, if anything, the main analysis may understate the magnitude of the effects, even if the instrument used does not fully address endogeneity.

C.4. Excluding S&P 500 additions and deletions

As discussed in Subsection II.C, local effects from my identification strategy involve companies owned by shareholders with more stable portfolios over time, as they would have a stronger first stage in 2SLS estimation. This subsection extends that intuition to companies with more stable ownership structures by excluding companies participating in, or related to, S&P 500 additions and deletions. I begin by discussing the literature on index reconstitutions to then focus specifically on the S&P 500 index. Next, I show that companies entering and exiting the S&P 500 list of constituents do not drive the results in Subsection III.A.

Replicating the analysis without S&P 500 turnover can help refine the conclusions from the main results. Most notably, it can indicate whether my main findings are sensitive to companies with more unstable ownership structures, which would require

tailoring the interpretation. In addition, it can potentially explain why OLS estimates are slightly downward biased.

Both Freeman (2025) and Section C.3 suggest that OLS estimates are, if anything, downward biased—even after flexibly controlling for potential confounders. This pattern may not seem peculiar, but usual endogeneity claims assume a positive correlation between ownership overlap and the residual.

Index reconstitutions can be a suitable explanation for this finding: index entrants may attract overlapping shareholders while being less likely to keep or form new trading partnerships.[28]

Additions and deletions in the literature. A substantial body of literature documents the effects of index reconstitutions. Early work shows that declines in deleted stocks are larger and more persistent than addition gains (Chen et al., 2004); however, more recent papers document a *disappearing index effect*, as deleted stocks ultimately outperform added ones, driven by a prominent rebound in institutional ownership and liquidity (Petajisto, 2011; Kot et al., 2015).

The tight connection found between institutional portfolios and index constituents inspired several papers to use reconstitutions as exogenous sources of variation in ownership structures. For example, Boone and White (2015), Kennedy et al. (2017), Brooks et al. (2018), and Kostovetsky and Manconi (2020) use Russell Index reconstitutions, while Aghion et al. (2013) use S&P 500 additions. The premise is that indexed funds have become a widely used form of passive investing—with most institutional shareholders typically employing a diversified strategy that combines active and passive funds. This strategy benefits minority shareholders by aggregating funds and tracking a sufficiently diversified set of assets that closely matches the composition and weights of indices and ETFs. Therefore, when index constituents change, passive funds—and other investors that replicate the strategy—must acquire equity in companies entering the index and sell stocks of companies leaving it. Even if index reconstitutions impose real costs on trackers (Madhavan, 2003; Petajisto, 2011), comovements in stocks suggest a correlated demand from index-tracking investors (DeCoste, 2025), which supports the intuition behind the mechanism.

However, Lewellen and Lowry (2021) argue that addressing the endogeneity of ownership structures is an inappropriate use of additions and deletions. First, which companies make the cut into the index depends on a committee, whose decisions might reflect the companies' most recent performance.[29] For instance, committee members could be anticipating entering companies to grow, or the opposite for companies leaving the index. Second, being added to the index attracts the attention of the media, analysts, and a wider range of investors. As a result, ownership structures can change through hard-to-isolate channels; for example, recent index additions see institutional ownership rise while blockholders are crowded out.

Boller and Morton (2020) innovates the literature on common ownership by proposing a different approach. Instead of focusing on the outcomes of the added companies, their strategy uses additions to study the outcomes of competitors, for whom the changes in

[28] Another potential, but untestable, channel runs through private-equity fund strategy. Investors such as Blackstone or KKR frequently acquire stakes in multiple distressed firms at once, spanning multiple industries. While this increases OSS among them, the same distress often reduces companies' propensity to keep or enter new trading partnerships. This generates a small downward bias in OLS, consistent with the modest gap between OLS and 2SLS estimates.

[29] According to the S&P 500 US Indices Methodology documentation: *"Constituent selection is at the discretion of the Index Committee and is based on the eligibility criteria. [...] Sector balance, [...] in the relevant market capitalization range, is also considered in the selection of companies for the indices."* It later adds, *"S&P Dow Jones Indices Index Committees reserve the right to make exceptions when applying the methodology if the need arises. In any scenario where the treatment differs from the general rules stated in this document or supplemental documents, clients will receive sufficient notice, whenever possible."*

index constituents are orthogonal. Studies such as Antón et al. (2023) and Azar et al. (2025) have later on popularized this empirical strategy.

S&P 500 reconstitutions. The S&P 500 index belongs to the *float-adjusted market-capitalization-weighted indices* segment of S&P Dow Jones, which has two immediate implications. First, market capitalization is based solely on floating outstanding shares. That is, it excludes shares held by long-term strategic shareholders, such as insiders, private equity firms, or the government. Second, stocks with higher market capitalization have a greater impact on the index's performance than those with lower market capitalization.

Thus, each company in the S&P 500 index has a weight that reflects its relative contribution to the index's total market capitalization. Therefore, when the S&P 500's constituents change, index weights shift proportionally, driven by differences in market value between additions and deletions. For example, when additions have a higher market capitalization than deletions, the index weights of all other constituents must fall to make room for the larger weights associated with the entrants. The opposite holds as well.

Because the index lists the 500 most influential companies in the US according to S&P Indices, a typical change in constituents should involve replacing a low-market-capitalization company with a high-market-capitalization one. However, this is seldom the case. Publicly listed companies in the US regularly engage in mergers, acquisitions, and spin-offs, prompting the S&P Indices Committee to respond quickly—so the index does not fall behind. As a result, the index does not always delete a company with a relatively low market value compared to the addition, making it challenging to anticipate the net change in the constituent companies' market values.[30]

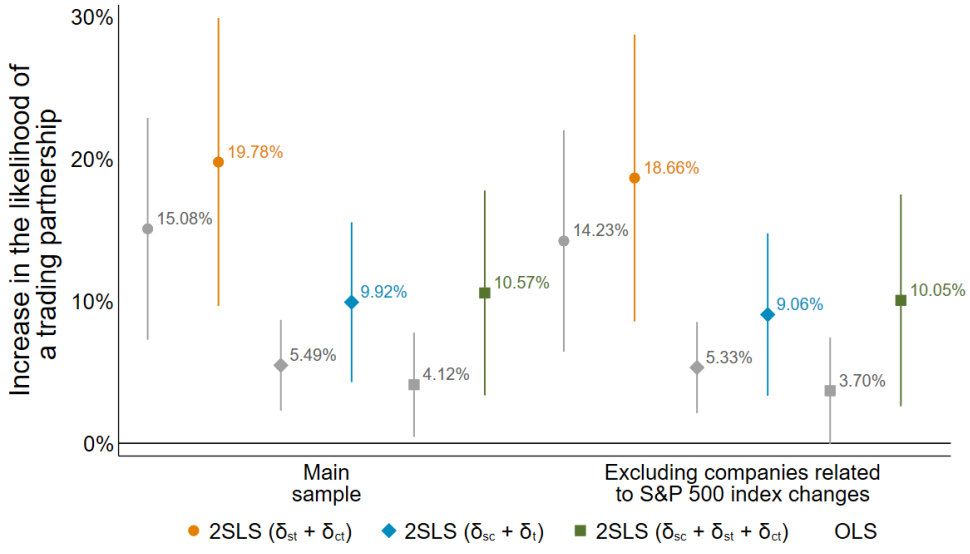
In any case, S&P 500 index changes constitute a predictable source of variation in ownership structures across US publicly listed companies. The empirical exercise in this subsection aims to show that fostering effects do not arise from sudden shifts in ownership overlap due to additions and deletions, by excluding not only entrant and exiting firms, but also companies that are directly related to the index change, such as parent companies, subsidiaries, spin-offs, or acquirers.

Robustness. For this robustness exercise, I manually collected press releases from Standard & Poor's, S&P Indices, and S&P Dow Jones Indices from 1999 to 2013, specifically those announcing additions to or deletions from the S&P 500. For each announcement, I identify the exact date of each addition and deletion, the names and tickers of the companies involved, and a detailed description of the justification behind the index change: a decline or rise in a company's market capitalization, a merger, an acquisition, a spin-off, a renaming, or a delisting.

This process allows me to identify not only entrants and exits to the S&P 500 index, but also parent companies, subsidiaries, spin-offs, acquirers, and other companies explicitly mentioned in the announcement. Notably, each explicit mention of a company in the press release is justified, showing that the Index Committee considered the relationship when deciding whether to add or delete a stock from the index.

[30] For example, on December 14, 2011, S&P Indices announced that *TripAdvisor Inc.* would replace *Tellabs Inc.* in the S&P 500 index by December 20, 2011. According to the press release, the announced day corresponds to the expected date on which the S&P 500 constituent Expedia was to complete the spin-off of TripAdvisor. Assuming a fixed number of outstanding shares and their prices, Expedia's market capitalization before the spin-off should equal the sum of TripAdvisor's and Expedia's during the first quarter of 2012. Therefore, this change in constituents should only reflect the market value lost at Tellabs' exit. Conversely, on March 27, 2000, Standard & Poor's announced that *Linear Technology Corp.* and *Pharmacia Corp.* would replace *Monsanto Company* and *Pharmacia & Upjohn* in the S&P 500 index. The press release explains that Pharmacia is the merger of Monsanto and Pharmacia & Upjohn. Therefore, the change in constituents solely reflects the added market value from Linear Technology's entry. There are many other examples, with 51 spin-offs, 158 mergers and acquisitions, and several name and ticker changes between 1999 and 2013.

Figure C.4 – *Conclusions are robust to excluding companies participating in, or related to, S&P 500 index changes*



Notes: The figure depicts the effect of a one-standard-deviation increase in *ownership-structure similarity* (OSS) on the average likelihood of a trading partnership with and without excluding companies participating in, or related to, S&P 500 index changes. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I define OSS in Equation (1). I estimate Equation (2) using OLS (left, gray) and 2SLS (right, color) across three FE specifications, as described in Subsection II.A: (customer×year)-FE and (supplier×year)-FE (orange circle), dyad-FE and year-FE (blue diamond), and both combined (green square). The 2SLS instrument is an analogous version of OSS, based on predicted ownership structures, as described in Subsection II.C. Confidence intervals report dyad-cluster robust standard errors at the 95% confidence level. All regressions drop singleton observations in each sample under the most stringent FE specification (green square). For further reference, see Subsection C.4 for a description of the analysis and Table SA.9 in the Appendix for details.

Next, for each press release, I drop from the sample all dyads involving one of the mentioned companies, but only for the year the addition or deletion took place and the following year. For example, *Sara Lee Corporation* announced in January, 2011 that the company would be split into two units. In June 2012, S&P Indices deleted the company from the S&P 500 index and added the post-spin stub to the S&P MidCap 400. Lastly, Sara Lee Corporation became delisted in July 2012, when the split became official. For the robustness exercise, I drop every dyad involving Sara Lee Corporation in 2012 and 2013. Note that I drop dyads for the following year, not only the year in which the addition or deletion takes place. This is to be consistent with my identification strategy, which leverages persistence in shareholdings from one year to the next.

I estimate Equation (2) for the resulting sample, which represents roughly 96% of the original observations, across the three fixed-effects specifications detailed in Subsection II.A, using OLS and 2SLS. For details, see Table SA.9 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages.

Figure C.4 compares coefficients across samples with and without companies related to S&P 500 index changes. Differences in estimates are statistically and numerically indistinguishable from zero, implying that additions and deletions cannot explain the downward bias in OLS estimates. In addition, this result suggests that sudden shifts in shareholdings do not drive the results, strengthening the main analysis's conclusion. That is, fostering effects involve shareholders with portfolios that experience few perturbations over time and, likely, companies with more stable ownership structures.

SUPPLEMENTARY MATERIAL

SA. Additional tables and figures

Table SA.1 – *Ownership-structure similarity (OSS) increases the likelihood of trading partnerships*

Dependent variable: Trading partnership (Avg. likelihood: 0.0203%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
OSS	0.1508***	0.0549***	0.0412*				0.1978***	0.0992***	0.1057**
	(0.039834)	(0.016272)	(0.018758)				(0.051663)	(0.028684)	(0.036771)
OSS:AR(1)				0.6932***	0.5248***	0.4507***			
				(0.000166)	(0.000200)	(0.000214)			
N	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088
F-statistic							17,482,474	6,879,157	4,435,221
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. OSS: 0.0926 (0.1369)									

Notes: The table reports OLS, first-stage (FS), and 2SLS estimates of the effect of a one-standard-deviation increase in OSS on the average likelihood of a trading partnership. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I define OSS in Equation (1). I estimate Equation (2) across three FE specifications, as described in Subsection II.A. The 2SLS instrument—OSS:AR(1)—is an analogous version of OSS, based on predicted ownership structures, as described in Subsection II.C. The table reports the Kleibergen-Paap rk Wald F-statistic and dyad-cluster robust standard errors in parentheses. All regressions drop singleton observations under the most stringent FE specification. For further reference, see Subsection III.A for a description of the analysis. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table SA.2 – *Relative shareholder concentration (RSC) amplifies the effect of ownership-structure similarity (OSS)*

Dependent variable: Trading partnership (Avg. likelihood: 0.0194%)

	OLS	FS			2SLS	OLS	FS			2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
OSS	0.3473***				0.4327***	0.1120***				0.1669***
	(0.066766)				(0.092181)	(0.028270)				(0.046134)
RSC	0.1920**				0.2236*	0.0278				0.0136
	(0.074325)				(0.102901)	(0.032560)				(0.065830)
OSS× RSC	0.3712***				0.4352**	0.1017*				0.1209
	(0.087071)				(0.136447)	(0.042749)				(0.079074)
OSS:AR(1)		0.7232***	-0.0395***	-0.1687***			0.5801***	-0.0239***	-0.1473***	
		(0.000207)	(0.000101)	(0.000225)			(0.000238)	(0.000113)	(0.000196)	
RSC:AR(1)		0.0238***	0.2431***	0.0121***			0.0562***	0.1229***	0.0182***	
		(0.000141)	(0.000213)	(0.000188)			(0.000148)	(0.000110)	(0.000122)	
OSS× RSC:AR(1)		0.0497***	-0.0008***	0.2145***			0.0841***	0.0079***	0.1414***	
		(0.000224)	(0.000128)	(0.000286)			(0.000252)	(0.000120)	(0.000219)	
N	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486
F-statistic		1,096,108	1,496,108	859,111	318,848		856,803	1,390,531	921,901	369,106
Company × Time FE	✓	✓	✓	✓	✓					
Dyad FE						✓	✓	✓	✓	✓
Time FE						✓	✓	✓	✓	✓
Avg. OSS: 0.1087 (0.1428) Avg. RSC: 0.7800 (0.9365)										

Notes: (*Continued on next page.*)

Table SA.2 – *Relative shareholder concentration (RSC) amplifies the effect of ownership-structure similarity (OSS)* (continued...)

Dependent variable: Trading partnership (Avg. likelihood: 0.0194%)

	OLS	FS			2SLS
	(11)	(12)	(13)	(14)	(15)
OSS	0.0921**				0.1864***
	(0.029801)				(0.056602)
RSC	-0.0296				-0.0194
	(0.039652)				(0.083329)
OSS× RSC	0.1056*				0.1397
	(0.043974)				(0.096101)
OSS:AR(1)		0.4972***	-0.0247***	-0.1236***	
		(0.000247)	(0.000101)	(0.000212)	
RSC:AR(1)		0.0272***	0.1576***	0.0037***	
		(0.000142)	(0.000180)	(0.000154)	
OSS× RSC:AR(1)		0.0613***	-0.0083***	0.1323***	
		(0.000205)	(0.000109)	(0.000212)	
N	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486
F-statistic		806,497	923,218	611,425	198,363
Company × Time FE	✓	✓	✓	✓	✓
Dyad FE	✓	✓	✓	✓	✓
Time FE					
Avg. OSS: 0.1087 (0.1428) Avg. RSC: 0.7800 (0.9365)					

Notes: The table reports OLS, first-stage (FS), and 2SLS estimates of the effects of one-standard-deviation increases in OSS, RSC, and their interaction on the likelihood of a trading partnership. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I keep dyads with RSC within one standard deviation of the mean, as described in Subsection II.A. I define OSS and RSC in Equation (1). I estimate Equation (3) across three FE specifications, as described in Subsection II.A. The 2SLS instruments—OSS:AR(1), RSC:AR(1), and OSS×RSC:AR(1)—are analogous versions of OSS, RSC, and their interaction, based on predicted ownership structures, as described in Subsection II.C. The table reports the Sanderson-Windmeijer F-statistics (FS), the Kleibergen-Paap rk Wald F-statistic (2SLS), and dyad-cluster robust standard errors in parentheses. All regressions drop singleton observations under the most stringent FE specification. For further reference, see Subsection III.B for a description of the analysis. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table SA.3 – *Tunneling effects are stronger when customers have more diffuse ownership*

Dependent variable: Trading partnership (Avg. likelihood: 0.0205%)

	Sub-sample: Dyads with diffuse-ownership suppliers					Sub-sample: Dyads with diffuse-ownership customers				
	OLS	FS			2SLS	OLS	FS			2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
OSS	0.0725				0.0676	0.5839***				0.7547***
	(0.058880)				(0.091128)	(0.121426)				(0.162328)
RSC										
OSS× RSC	-0.0671				-0.1706	0.7732***				0.9537***
	(0.090129)				(0.151046)	(0.157009)				(0.228498)
OSS:AR(1)		0.7029***		-0.1635***			0.7026***		-0.1609***	
		(0.000312)		(0.000343)			(0.000314)		(0.000344)	
RSC:AR(1)										
OSS× RSC:AR(1)		0.0313***		0.2566***			0.0283***		0.2611***	
		(0.000370)		(0.000510)			(0.000370)		(0.000506)	
N	55,481,190	55,481,190		55,481,190	55,481,190	55,482,039	55,482,039		55,482,039	55,482,039
F-statistic		557,866		316,209	151,608		560,997		321,527	152,562
Company × Time FE	✓	✓		✓	✓	✓	✓		✓	✓
Dyad FE										
Time FE										
Avg. OSS: 0.1180 (0.1468) Avg. RSC: 0.7800 (0.9365)										

Notes: (*Continued on next page.*)

Table SA.3 – *Tunneling effects are stronger when customers have more diffuse ownership* (continued...)

Dependent variable: Trading partnership (Avg. likelihood: 0.0205%)

	Sub-sample: Dyads with diffuse-ownership suppliers					Sub-sample: Dyads with diffuse-ownership customers				
	OLS	FS			2SLS	OLS	FS			2SLS
	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
OSS	-0.0402				-0.0801	0.2631***				0.4969***
	(0.027171)				(0.056024)	(0.051117)				(0.101193)
RSC	-0.0618				-0.1330	0.0967				0.3282*
	(0.033972)				(0.104915)	(0.055455)				(0.133000)
OSS× RSC	-0.0707				-0.1721	0.2173**				0.4791**
	(0.041649)				(0.107647)	(0.069645)				(0.155861)
OSS:AR(1)		0.5197***	-0.0182***	-0.1392***			0.5182***	-0.0162***	-0.1375***	
		(0.000356)	(0.000171)	(0.000279)			(0.000356)	(0.000171)	(0.000282)	
RSC:AR(1)		0.0257***	0.1158***	0.0232***			0.0229***	0.1160***	0.0261***	
		(0.000243)	(0.000179)	(0.000209)			(0.000240)	(0.000174)	(0.000208)	
OSS× RSC:AR(1)		0.0318***	0.0115***	0.1400***			0.0278***	0.0134***	0.1434***	
		(0.000371)	(0.000194)	(0.000341)			(0.000368)	(0.000192)	(0.000338)	
N	55,481,190	55,481,190	55,481,190	55,481,190	55,481,190	55,482,039	55,482,039	55,482,039	55,482,039	55,482,039
F-statistic		269,780	504,964	310,175	133,450		269,887	514,972	317,982	134,049
Company × Time FE										
Dyad FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Avg. OSS: 0.1180 (0.1468) Avg. RSC: 0.7800 (0.9365)									

Notes: The table reports OLS, first-stage (FS), and 2SLS estimates of the effects of one-standard-deviation increases in *ownership-structure similarity* (OSS), *relative shareholder concentration* (RSC), and their interaction on the likelihood of a trading partnership, depending on whether customers or suppliers have a relatively diffuse ownership structure. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I split the analysis by the side of the trading partnership with the more diffuse ownership structure. I keep dyads with RSC within one standard deviation of the mean, as described in Subsection II.A. I define OSS and RSC in Equation (1). (*Continued on next page.*)

Table SA.3 – *Tunneling effects are stronger when customers have more diffuse ownership* (continued...)

Dependent variable: Trading partnership (Avg. likelihood: 0.0205%)

	Sub-sample: Dyads with diffuse-ownership suppliers					Sub-sample: Dyads with diffuse-ownership customers				
	OLS	FS			2SLS	OLS	FS			2SLS
	(21)	(22)	(23)	(24)	(25)	(26)	(27)	(28)	(29)	(30)
OSS	-0.0427				-0.0796	0.2060***				0.4341***
	(0.033343)				(0.084216)	(0.053756)				(0.114567)
RSC										
OSS× RSC	-0.0813				-0.2865*	0.2431**				0.4940**
	(0.043930)				(0.142040)	(0.074910)				(0.187540)
OSS:AR(1)		0.4431***		-0.1143***			0.4425***		-0.1126***	
		(0.000374)		(0.000319)			(0.000376)		(0.000320)	
RSC:AR(1)										
OSS× RSC:AR(1)		0.0207***		0.1391***			0.0184***		0.1419***	
		(0.000338)		(0.000375)			(0.000338)		(0.000374)	
N	55,481,190	55,481,190		55,481,190	55,481,190	55,482,039	55,482,039		55,482,039	55,482,039
F-statistic		223,688		178,037	82,705		225,390		179,596	82,911
Company × Time FE	✓	✓		✓	✓	✓	✓		✓	✓
Dyad FE	✓	✓		✓	✓	✓	✓		✓	✓
Time FE										
Avg. OSS: 0.1180 (0.1468) Avg. RSC: 0.7800 (0.9365)										

Notes: I estimate Equation (3) separately on each subsample, across three FE specifications, as described in Subsection II.A. The 2SLS instruments—OSS:AR(1), RSC:AR(1), and OSS×RSC:AR(1)—are analogous versions of OSS, RSC, and their interaction, based on predicted ownership structures, as described in Subsection II.C. The table reports the Sanderson-Windmeijer F-statistics (FS), the Kleibergen-Paap rk Wald F-statistic (2SLS), and dyad-cluster robust standard errors in parentheses. All regressions drop singleton observations in each subsample under the most stringent FE specification. Within each subsample, RSC is collinear with the Dyad FE and is therefore dropped from the estimation. For further reference, see Subsection III.B for a description of the analysis. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table SA.4 – *Vertically overlapping ownership alleviates holdup*

Dependent variable: Trading partnership (Avg. likelihood: 0.0200%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A — $R\&D^{Low}_{SUP}$ &$R\&D^{Low}_{CUS}$									
OSS	0.1046	0.0566*	0.0393				0.1433	0.1009*	0.1068*
	(0.061990)	(0.023920)	(0.027538)				(0.078830)	(0.040401)	(0.051872)
OSS:AR(1)				0.7080***	0.5470***	0.4708***			
				(0.000225)	(0.000269)	(0.000293)			
N	73,904,798	73,904,798	73,904,798	73,904,798	73,904,798	73,904,798	73,904,798	73,904,798	73,904,798
F-statistic							9,928,030	4,128,420	2,573,284
Panel B — $R\&D^{Low}_{SUP}$ &$R\&D^{High}_{CUS}$									
OSS	-0.0148	0.0266	0.0344				-0.0275	0.0519	0.0688
	(0.029359)	(0.017487)	(0.021335)				(0.040149)	(0.030954)	(0.042832)
OSS:AR(1)				0.6719***	0.5017***	0.4298***			
				(0.000366)	(0.000440)	(0.000473)			
N	35,648,430	35,648,430	35,648,430	35,648,430	35,648,430	35,648,430	35,648,430	35,648,430	35,648,430
F-statistic							3,366,960	1,302,956	825,772
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. OSS: 0.0931 (0.1372)									

Notes: The table reports OLS, first-stage (FS), and 2SLS estimates of the effect of a one-standard-deviation increase in *ownership-structure similarity* (OSS) on the average likelihood of a trading partnership across four settings representing different degrees of holdup. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I fix each company's R&D intensity in the first year I observe a non-missing value and keep only years after that first observation. If a company has no non-missing value between 1999 and 2012, I impute the median value within 4-digit SIC industries. I classify customers and suppliers above or below the average R&D intensity among the remaining firms in the sample, excluding the top decile (8.30%). (*Continued on next page.*)

Table SA.4 – *Vertically overlapping ownership alleviates holdup* (continued...)

Dependent variable: Trading partnership (Avg. likelihood: 0.0200%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel C — $R\&D_{SUP}^{High}$ $\&R\&D_{CUS}^{Low}$									
OSS	0.4092***	0.0869*	0.0636				0.5191***	0.1489	0.1566
	(0.099125)	(0.043832)	(0.054249)				(0.132276)	(0.080443)	(0.109241)
OSS:AR(1)				0.6731***	0.5032***	0.4309***			
				(0.000363)	(0.000428)	(0.000463)			
N	35,649,661	35,649,661	35,649,661	35,649,661	35,649,661	35,649,661	35,649,661	35,649,661	35,649,661
F-statistic							3,430,675	1,378,944	865,139
Panel D — $R\&D_{SUP}^{High}$ $\&R\&D_{CUS}^{High}$									
OSS	0.3650***	0.0163	-0.0055				0.5143***	0.0353	0.0219
	(0.096574)	(0.049516)	(0.056679)				(0.132706)	(0.095072)	(0.122865)
OSS:AR(1)				0.6485***	0.4613***	0.3999***			
				(0.000576)	(0.000672)	(0.000729)			
N	17,180,427	17,180,427	17,180,427	17,180,427	17,180,427	17,180,427	17,180,427	17,180,427	17,180,427
F-statistic							1,266,765	471,746	301,301
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. OSS: 0.0931 (0.1372)									

Notes: Then, I split the sample, assigning each dyad *sc* to one of four possible subsamples $g(sc) \in \{\text{LL, LH, HL, HH}\}$. I define OSS in Equation (1). I estimate Equation (B.1) separately on each subsample, across three FE specifications, as described in Subsection II.A. The 2SLS instrument—OSS:AR(1)—is an analogous version of OSS, based on predicted ownership structures, as described in Subsection II.C. The table reports the Kleibergen-Paap rk Wald F-statistic and dyad-cluster robust standard errors in parentheses. All regressions drop singleton observations in each subsample under the most stringent FE specification. For further reference, see Subsection B.1 for a description of the analysis. *** p < 0.001, ** p < 0.01, * p < 0.05.

Table SA.5 – *Vertically overlapping ownership alleviates double marginalization*

Dependent variable: Trading partnership (Avg. likelihood, pooled: 0.0189%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A — HHI_{SUP}^{Low} &HHI_{CUS}^{Low}									
OSS	0.2874**	0.1181*	0.0888				0.3853**	0.2298*	0.2471
	(0.104545)	(0.055526)	(0.074258)				(0.135495)	(0.099433)	(0.148332)
OSS:AR(1)				0.6844***	0.5216***	0.4448***			
				(0.000373)	(0.000447)	(0.000473)			
N	26,428,564	26,428,564	26,428,564	26,428,564	26,428,564	26,428,564	26,428,564	26,428,564	26,428,564
F-statistic							3,370,000	1,359,910	885,384
Panel B — HHI_{SUP}^{Low} &HHI_{CUS}^{High}									
OSS	0.0495	0.0068	0.0039				0.0549	0.0189	0.0449
	(0.078761)	(0.036827)	(0.042301)				(0.104168)	(0.062114)	(0.075443)
OSS:AR(1)				0.6972***	0.5408***	0.4655***			
				(0.000339)	(0.000387)	(0.000423)			
N	35,524,266	35,524,266	35,524,266	35,524,266	35,524,266	35,524,266	35,524,266	35,524,266	35,524,266
F-statistic							4,236,746	1,952,579	1,208,645
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. OSS (pooled): 0.0990 (0.140235)									

Notes: The table reports OLS, first-stage (FS), and 2SLS estimates of the effect of a one-standard-deviation increase in *ownership-structure similarity* (OSS) on the average likelihood of a trading partnership across four settings representing different degrees of double marginalization. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I fix each company's HHI-TNIC in the first year I observe a non-missing value and keep only years after that first observation. I classify customers and suppliers above or below 1,500 HHI-TNIC, which corresponds to the threshold that separates *unconcentrated* and *moderately concentrated* markets, and then split the sample, assigning each dyad sc to one of four possible subsamples $h(sc) \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\}$. (*Continued on next page.*)

Table SA.5 – *Vertically overlapping ownership alleviates double marginalization* (continued...)

Dependent variable: Trading partnership (Avg. likelihood, pooled: 0.0189%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel C — HHI_{SUP}^{High} &HHI_{CUS}^{Low}									
OSS	0.0797	0.0239	0.0582				0.1519*	0.0461	0.1506*
	(0.060843)	(0.027433)	(0.032561)				(0.074252)	(0.049858)	(0.063660)
OSS:AR(1)				0.6966***	0.5395***	0.4655***			
				(0.000335)	(0.000383)	(0.000418)			
N	35,524,229	35,524,229	35,524,229	35,524,229	35,524,229	35,524,229	35,524,229	35,524,229	35,524,229
F-statistic							4,323,690	1,982,176	1,239,393
Panel D — HHI_{SUP}^{High} &HHI_{CUS}^{High}									
OSS	0.2140*	0.1221***	0.0669				0.2688*	0.1923***	0.1497*
	(0.097305)	(0.033570)	(0.036831)				(0.121421)	(0.054654)	(0.067832)
OSS:AR(1)				0.7095***	0.5518***	0.4829***			
				(0.000290)	(0.000328)	(0.000366)			
N	47,740,511	47,740,511	47,740,511	47,740,511	47,740,511	47,740,511	47,740,511	47,740,511	47,740,511
F-statistic							5,980,651	2,827,327	1,736,939
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. OSS (pooled): 0.0990 (0.140235)									

Notes: I define OSS in Equation (1). I estimate Equation (B.2) separately on each subsample, across three FE specifications, as described in Subsection II.A. The 2SLS instrument—OSS:AR(1)—is an analogous version of OSS, based on predicted ownership structures, as described in Subsection II.C. The table reports the Kleibergen-Paap rk Wald F-statistic and dyad-cluster robust standard errors in parentheses. All regressions drop singleton observations in each subsample under the most stringent FE specification. For further reference, see Subsection B.2 for a description of the analysis. *** p < 0.001, ** p < 0.01, * p < 0.05.

Table SA.6 – *Alternative sample definitions do not overturn the effect of ownership-structure similarity (OSS)*

Dependent variable: Trading partnership

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A — Main analysis									
OSS	0.1508***	0.0549***	0.0412*				0.1978***	0.0992***	0.1057**
	(0.039834)	(0.016272)	(0.018758)				(0.051663)	(0.028684)	(0.036771)
OSS:AR(1)				0.6932***	0.5248***	0.4507***			
				(0.000166)	(0.000200)	(0.000214)			
N	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088
F-statistic							17,482,474	6,879,157	4,435,221
	Avg. OSS: 0.0926 (0.1369) Avg. likelihood: 0.0203%								
Panel B — Feasible dyads									
OSS	0.1831***	0.0784***	0.0488*				0.2387***	0.1383***	0.1247**
	(0.046167)	(0.018719)	(0.021723)				(0.059874)	(0.033000)	(0.042608)
OSS:AR(1)				0.6932***	0.5248***	0.4507***			
				(0.000167)	(0.000203)	(0.000216)			
N	164,847,516	164,847,516	164,847,516	164,847,516	164,847,516	164,847,516	164,847,516	164,847,516	164,847,516
F-statistic							17,230,559	6,697,452	4,369,500
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
	Avg. OSS: 0.0926 (0.1369) Avg. likelihood: 0.0175%								

Notes: (*Continued on next page.*)

Table SA.6 – *Alternative sample definitions do not overturn the effect of ownership-structure similarity (OSS)* (continued...)

Dependent variable: Trading partnership

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel C — Fully-weighted partnerships									
OSS	0.0683***	0.0009	0.0012*				0.0912***	0.0009	0.0020
	(0.013591)	(0.000476)	(0.000556)				(0.017613)	(0.000832)	(0.001083)
OSS:AR(1)				0.6959***	0.5237***	0.4501***			
				(0.000306)	(0.000326)	(0.000292)			
N	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088
F-statistic							5,163,283	2,573,414	2,374,504
	Avg. OSS: 0.0938 (0.1384) Avg. likelihood: 2.3678%								
Panel D — Historical supply chain									
OSS	0.1581***	0.1293***	0.0318*				0.2176***	0.1657***	0.0691*
	(0.025226)	(0.012728)	(0.015766)				(0.031771)	(0.020303)	(0.029003)
OSS:AR(1)				0.7441***	0.6039***	0.5217***			
				(0.000554)	(0.000663)	(0.000731)			
N	5,930,597	5,930,597	5,930,597	5,930,597	5,930,597	5,930,597	5,930,597	5,930,597	5,930,597
F-statistic							1,802,470	828,394	509,762
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
	Avg. OSS: 0.0943 (0.1507) Avg. likelihood: 0.4555%								

Notes: (*Continued on next page.*)

Table SA.6 – *Alternative sample definitions do not overturn the effect of ownership-structure similarity (OSS)* (continued...)

Dependent variable: Trading partnership

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel E — TNIC-based competitors									
OSS	0.0913***	0.0203	0.0152				0.1191***	0.0444	0.0567
	(0.022847)	(0.012270)	(0.014308)				(0.030429)	(0.024877)	(0.034927)
OSS:AR(1)				0.7047***	0.4685***	0.3872***			
				(0.001076)	(0.001591)	(0.001698)			
N	1,432,444	1,432,444	1,432,444	1,432,444	1,432,444	1,432,444	1,432,444	1,432,444	1,432,444
F-statistic							429,253	86,735	52,005
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. OSS: 0.1147 (0.1608) Avg. likelihood: 1.7399%									

Notes: The table reports OLS, first-stage (FS), and 2SLS estimates of the effect of a one-standard-deviation increase in OSS on the average likelihood of a trading partnership across five sample definitions, based on feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I define OSS in Equation (1). I estimate Equation (2) on each sample, across three FE specifications, as described in Subsection II.A. The 2SLS instrument—OSS:AR(1)—is an analogous version of OSS, based on predicted ownership structures, as described in Subsection II.C. The table reports the Kleibergen-Paap rk Wald F-statistic and dyad-cluster robust standard errors in parentheses. All regressions drop singleton observations in each sample under the most stringent FE specification. For further reference, see Subsection C.1 for a description of the analysis. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table SA.7 – *Alternative overlapping-ownership metrics do not overturn the effect of ownership-structure similarity (OSS)*

Dependent variable: Trading partnership (Avg. likelihood: 0.0203%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A — Ownership-structure similarity (OSS)									
OSS	0.1509***	0.0549***	0.0412*				0.1979***	0.0990***	0.1059**
	(0.039860)	(0.016281)	(0.018773)				(0.051699)	(0.028701)	(0.036810)
OSS:AR(1)				0.6933***	0.5248***	0.4507***			
				(0.000167)	(0.000202)	(0.000215)			
N	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470
F-statistic							17,297,004	6,735,506	4,378,352
Avg. OSS: 0.0926 (0.136929)									
Panel B — Overlapping-shares cross-product (OSCP)									
OSCP	0.2627***	0.1108***	0.0585				0.3137***	0.2652***	0.1542*
	(0.054158)	(0.027412)	(0.034024)				(0.095138)	(0.044908)	(0.064937)
OSCP:AR(1)				0.9778***	0.8567***	0.7694***			
				(0.000771)	(0.000343)	(0.000390)			
N	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470
F-statistic							1,606,784	6,252,289	3,900,807
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. OSCP: 0.0547 (0.092987)									

Notes: (*Continued on next page.*)

Table SA.7 – *Alternative overlapping-ownership metrics do not overturn the effect of ownership-structure similarity (OSS)* (continued...)

Dependent variable: Trading partnership (Avg. likelihood: 0.0203%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel C — Overlapping market value (OMV)									
OMV	0.0366	-0.0159	-0.0610				0.1229	0.0889*	-0.0042
	(0.051204)	(0.023450)	(0.033950)				(0.079439)	(0.037076)	(0.061734)
OMV:AR(1)				0.8937***	0.7268***	0.6322***			
				(0.000178)	(0.000199)	(0.000205)			
N	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470
F-statistic							25,155,483	13,365,227	9,530,321
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. OMV: 0.1610 (0.172495)									

Notes: The table reports OLS, first-stage (FS), and 2SLS estimates of the effect of a one-standard-deviation increase in ownership overlap on the average likelihood of a trading partnership across three definitions of overlapping ownership. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I define OSS in Equation (1); OSCP and OMV in Subsection C.2. I estimate Equation (2) using 2SLS across three FE specifications, as described in Subsection II.A. The 2SLS instrument—OSS:AR(1), OSCP:AR(1), and OMV:AR(1)—are analogous versions of OSS, OSCP, and OMV, respectively, based on predicted ownership structures, as described in Subsection II.C. The table reports the Kleibergen-Paap rk Wald F-statistic and dyad-cluster robust standard errors in parentheses. All regressions drop singleton observations under the most stringent FE specification. For further reference, see Subsection C.2 for a description of the analysis.
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table SA.8 – *Alternative prediction models for the instrument do not overturn the effect of ownership-structure similarity (OSS)*

Dependent variable: Trading partnership (Avg. likelihood: 0.0214%)

	OLS	AR(1) model		AR(2) model		Owner-specific AR(1,k) model	
	(1) OLS	(2) FS	(3) 2SLS	(4) FS	(5) 2SLS	(6) FS	(7) 2SLS
OSS	0.1514***		0.2014***		0.2581***		0.1948***
	(0.040111)		(0.051576)		(0.065317)		(0.052329)
OSS:AR(1)		0.6966***					
		(0.000166)					
OSS:AR(2)				0.4945***			
				(0.000212)			
OSS:AR(1,k)						0.6711***	
						(0.000170)	
F-statistic			17,518,434		5,418,522		15,565,732
Company × Time FE	✓	✓	✓	✓	✓	✓	✓
OSS	0.0517**		0.1054***		0.1973***		0.0902**
	(0.016531)		(0.028795)		(0.048355)		(0.028717)
OSS:AR(1)		0.5265***					
		(0.000209)					
OSS:AR(2)				0.2769***			
				(0.000219)			
OSS:AR(1,k)						0.4874***	
						(0.000200)	
F-statistic							
Dyad FE	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓
OSS	0.0405*		0.1031**		0.2886***		0.1055**
	(0.019549)		(0.037706)		(0.079628)		(0.037686)
OSS:AR(1)		0.4521***					
		(0.000223)					
OSS:AR(2)				0.1838***			
				(0.000218)			
OSS:AR(1,k)						0.4443***	
						(0.000224)	
F-statistic							
Company × Time FE	✓	✓	✓	✓	✓	✓	✓
Dyad FE	✓	✓	✓	✓	✓	✓	✓
N	142,691,158	142,691,158	142,691,158	142,691,158	142,691,158	142,691,158	142,691,158
Avg. OSS: 0.0990 (0.1405)							

Notes: The table reports OLS, first-stage (FS), and 2SLS estimates of the effect of a one-standard-deviation increase in OSS on the average likelihood of a trading partnership using three different instruments. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2001 and 2013, as described in Subsection I.B. I define OSS in Equation (1). I estimate Equation (2) across three FE specifications, as described in Subsection II.A. The 2SLS instruments—OSS:AR(1), OSS:AR(2), and OSS:AR(1,k)—are analogous versions of OSS, based on predicted ownership structures. Each instrument corresponds to a different empirical model, as described in Subsection II.C and C.3: a first-order Markovian process, AR(1); a single-coefficient second-order Markovian process, AR(2); and a first-order Markovian process with investor-specific parameters, AR(1,k). The table reports the Kleibergen-Paap rk Wald F-statistic and dyad-cluster robust standard errors in parentheses. All regressions drop singleton observations under the most stringent FE specification. For further reference, see Subsection C.3 for a description of the analysis.
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table SA.9 – *Conclusions are robust to excluding companies participating in, or related to, S&P 500 index changes*

Dependent variable: Trading partnership

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A — Main sample									
OSS	0.1508***	0.0549***	0.0412*				0.1978***	0.0992***	0.1057**
	(0.039834)	(0.016272)	(0.018758)				(0.051663)	(0.028684)	(0.036771)
OSS:AR(1)				0.6932***	0.5248***	0.4507***			
				(0.000166)	(0.000200)	(0.000214)			
N	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088
F-statistic							17,482,474	6,879,157	4,435,221
	Avg. OSS: 0.0926 (0.1369) Avg. likelihood: 0.0203%								
Panel B — Excluding companies participating in, or related to, S&P 500 index changes									
OSS	0.1423***	0.0533**	0.0370				0.1866***	0.0906**	0.1005**
	(0.039763)	(0.016368)	(0.019147)				(0.051479)	(0.029140)	(0.038004)
OSS:AR(1)				0.6915***	0.5252***	0.4502***			
				(0.000172)	(0.000207)	(0.000222)			
N	158,229,494	158,229,494	158,229,494	158,229,494	158,229,494	158,229,494	158,229,494	158,229,494	158,229,494
F-statistic							16,230,683	6,451,777	4,105,801
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
	Avg. OSS: 0.0907 (0.1351) Avg. likelihood: 0.0187%								

Notes: The table reports OLS, first-stage (FS), and 2SLS estimates of the effect of a one-standard-deviation increase in *ownership-structure similarity* (OSS) on the average likelihood of a trading partnership with and without excluding companies participating in, or related to, S&P 500 index changes. The sample consists of feasible customer-supplier dyads for publicly listed US companies between 2000 and 2013, as described in Subsection I.B. I define OSS in Equation (1). I estimate Equation (2), across three FE specifications, as described in Subsection II.A. The 2SLS instrument—OSS:AR(1)—is an analogous version of OSS, based on predicted ownership structures, as described in Subsection II.C. The table reports the Kleibergen-Paap rk Wald F-statistic and dyad-cluster robust standard errors in parentheses. All regressions drop singleton observations in each sample under the most stringent FE specification. For further reference, see Subsection C.4 for a description of the analysis. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.